Characterizing the Detectability Landscape of Iterative Model Collapse

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Abstract

When language models train on their own output, distributional quality degrades well before standard metrics register collapse. Detecting this degradation early is difficult: the time series are short ($T \leq 25$ generations), the signal is sub-linear in contamination fraction, and, as we show, the choice of statistical test can shift conclusions from universal detection to complete null. We map the detectability landscape of iterative model collapse, primarily on Pythia-410M across 13 contamination levels with cross-architecture validation on five families (Pythia, GPT-2, OPT, Qwen, SmolLM3; 345M–6.9B). Using permutation Granger causality[1] (PermG), we identify a detection boundary $\alpha_c(T)$ that follows a power law in observation length (leave-one-T-out CV: MAE$=0.037$, 15/15 cells within CI), structured into four regimes: a blind zone, a sharp onset, a non-monotonic dip, and a reliable detection region. Five statistical methods applied to identical data yield null-control false positive rates from 3% to 92%, exposing method sensitivity as a primary concern rather than a robustness footnote. Entropy Granger-precedes diversity collapse by ≥ 1 generation (14/15 experiments), providing a canary signal whose triggered cessation reduces peak perplexity by 64% in a proof-of-concept intervention.

1 Introduction

When language models train iteratively on their own output, the training distribution progressively loses its tail modes, a phenomenon termed model collapse (Shumailov et al., 2024; Alemohammad et al., 2024). Lexical diversity narrows, calibration deteriorates, and each generation inherits the systematic biases of its predecessors. As machine-generated text becomes prevalent on the web, such feedback loops pose a growing risk to production training pipelines. By the time standard quality metrics register degradation, the underlying distribution may have already shifted irreversibly.

This threat motivates a fundamental question: *what does the detectability landscape of model collapse look like?* At what contamination level does statistical detection become possible, how does this boundary depend on observation length, and what structure does the landscape exhibit? These questions are nontrivial because iterative training produces short, noisy time series (often fewer than 15 points), where detection power drops sharply (Boettiger and Hastings, 2012). Moreover, model collapse is a first-order transition (Truong and Truong, 2026), structurally ruling out the critical-slowing-down early-warning signals that ecologists rely on (Scheffer et al., 2009) (§2).

We map this landscape on Pythia-410M (Biderman et al., 2023) across 13 contamination levels with multi-seed replication, and find that it has rich structure (Figure 1). Rather than a smooth degradation curve, the landscape contains blind zones where collapse proceeds undetected, a sharp onset where detection jumps from null to reliable within a 2-percentage-point interval, a non-monotonic dip where detection temporarily fails, and a power-law detection boundary $\alpha_c(T) \propto T^{-2.08}$. Cross-architecture validation across five families (345M–6.9B) confirms that this structure generalizes when entropy-diversity coupling is preserved. A second finding is equally consequential: five statistical methods applied to identical data yield null-control false-positive rates from 3% to 92%, demonstrating that the landscape one observes depends critically on the method used to probe it. Granger precedence establishes temporal predictive improvement, not true causality (Shojaie and Fox, 2022); our canary claim is scoped to diversity metrics (distinct-n (Li et al., 2016)).

Our contributions are:

[1] Granger precedence establishes temporal predictive improvement, not true causation (Shojaie and Fox, 2022).

1. **Detectability landscape characterization** (§5.2–5.6). PermG establishes a detection boundary $\alpha_c(T)$ following a power law in observation length (Eq. 3; leave-one-T-out CV: MAE=0.037, 15/15 cells within CI). Any nonzero contamination produces signal, but sub-linear growth makes detection a function of T. Canary-triggered cessation reduces peak perplexity by 64%.

2. **Formal design rationale for canary metric selection** (§3). Under mode amplification, entropy responds before diversity by $\geq$1 generation (Design Principles 1–2), confirmed in 14/15 experiments ($\alpha \geq 0.30$; GPT-2-medium s43 bidirectional).

3. **Method sensitivity as a primary finding** (§4.1–5.3). Five detection methods yield null-control FPR from 3% to 92%; the same Granger test shifts from 0/8 to 6/8 within a 2-percentage-point interval, motivating method sensitivity as a first-order concern.

2 Related Work

Model collapse and canary metrics. Iterative training collapse was formalized by Shumailov et al. (2024) and Alemohammad et al. (2024), who showed that recursively replacing real data with model-generated text causes progressive tail erosion; Hataya et al. (2023) demonstrated analogous degradation in image generation; Gerstgrasser et al. (2024) demonstrated that data accumulation (rather than replacement) can mitigate collapse; Schaeffer et al. (2025) systematize eight distinct notions of "collapse" that this literature conflates. Our operationalization (declining detection rate of Granger-precedence canary→diversity links across generations) aligns most closely with Schaeffer et al.'s *distributional shift* family (definitions D4–D6 in their taxonomy), measuring how the generated distribution departs from the real-data baseline rather than tracking population risk or scaling-law exponents. Theoretical analyses have sharpened these results: Dohmatob et al. (2024) recast collapse as a change in scaling-law exponents, Dohmatob et al. (2025) proved that collapse can occur even at vanishingly small synthetic fractions, Seddik et al. (2025) provide a statistical analysis of iterative contamination dynamics, and Marchi et al. (2025) challenge the universality of collapse under diverse training conditions. Several works have proposed monitoring metrics: Du et al. (2025) report ECE increases at 30% contamination before accuracy drops, Gambetta et al. (2024) introduces surplexity as a collapse detector, Drayson et al. (2025) show that machine-generated text detection can prevent collapse, and Li et al. (2025) propose verification-based mitigation. Shi et al. (2025) independently validate entropy as a collapse diagnostic in diffusion models (r=0.91 with generalization score), using it for diversity-aware sample selection. Du and Tanaka-Ishii (2026) address a related but distinct phenomenon (mode collapse during generation) via geometric regulation of the representation trajectory; our work instead studies *iterative training* collapse across generations. We differ from prior monitoring work in four respects (comparison table: Appendix K): (i) we compare an entire family of canary metrics; (ii) we cross-validate across five statistical methods; (iii) we map detectability as a function of contamination regime; and (iv) we treat method-selection sensitivity itself as a primary finding.

Data mixing phase transitions. The observation that critical mixture ratios separate qualitatively different outcomes has precedent: Kazdan et al. (2025) map conditions under which synthetic data triggers collapse versus improves performance, and Gu et al. (2024) establish a scaling law for critical mixture ratios in continual pre-training. Gu et al. (2025) show that data mixing can induce phase transitions in knowledge acquisition, a direct theoretical analogue to our detection boundary $\alpha_c(T)$.

Critical transitions and lead-lag methodology. Scheffer et al. (2009) established that systems approaching tipping points exhibit generic early-warning signals (rising autocorrelation and variance); Dakos et al. (2012) developed a practical toolkit for estimating such signals, while Boettiger and Hastings (2012) quantified detection limits in short time series, directly relevant to iterative training where T=5–20. Truong and Truong (2026) prove that entropy collapse in iterative training is a first-order (discontinuous) phase transition, structurally ruling out critical-slowing-down (CSD) early-warning signals. Our Granger-precedence approach circumvents this limitation by detecting temporal predictive improvement rather than variance/autocorrelation signatures, achieving $\geq 5/8$ forward detection in 12 of 15 above-boundary cells (three sync-dip exceptions at α=0.60 dissolve by T=25; §5.4). The standard framework for testing

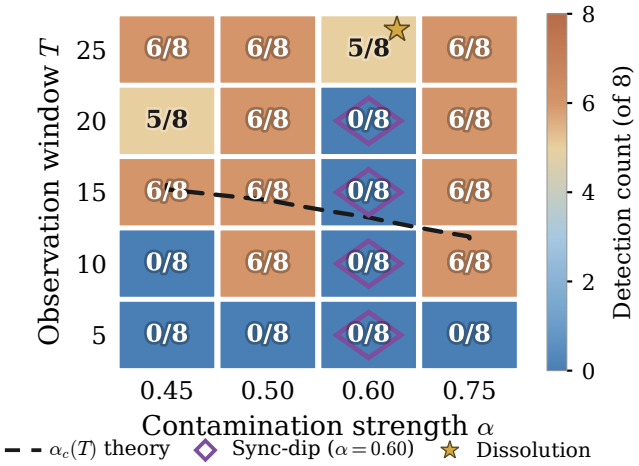


Figure 1: Detection phase diagram (Pythia-410M, fp32, PermG direction-checked). Each cell shows the number of FDR-significant canary→diversity pairs (of 8) at a given contamination strength α and observation window T. The dashed curve is the theoretical detection boundary $\alpha_c(T) \propto T^{-\gamma}$ with γ=2.08, fitted (Eq. 3). Diamond markers ($\diamond$) flag the synchronous dip at α=0.60 (0/8 at $T \leq 20$); the star ($\star$) marks its dissolution at T=25 (5/8), validating $\sqrt{T}$ power scaling.

temporal precedence is Granger causality (Granger, 1969), with neural extensions for nonlinear settings (Tank et al., 2021; Marcinkevics and Vogt, 2022; Zeng et al., 2025); however, non-stationary series inflate rejection rates (Granger and Newbold, 1974), and the Toda–Yamamoto procedure (Toda and Yamamoto, 1995) addresses this at reduced degrees of freedom severe in short series. We use classical rather than neural Granger methods because $T \leq 25$ provides insufficient observations for neural network parameter estimation. These methods are largely absent from the model collapse literature. Our analysis studies iterative from-scratch retraining; online or continual learning settings, where the model updates incrementally on streaming data, may exhibit different detection dynamics.

Analytical robustness. Bell et al. (2022) introduced multiverse analysis to ML, Coretta et al. (2023) demonstrated that 46 teams reached different conclusions on identical data, and Kapoor et al. (2024) synthesized these concerns into a consensus checklist. Our work applies this perspective to model collapse: five methods across multiple contamination regimes produce conclusions ranging from universal significance to complete null.

3 Cascade Ordering

If distributional collapse follows a predictable ordering across metrics, the earliest-responding metric can serve as a canary signal. We formalize known metric properties under mode amplification, the assumption that MLE on peaked distributions concentrates probability toward high-frequency tokens (Shumailov et al., 2024; Dohmatob et al., 2024) (derivations in Appendix X).

Design Principle 1 (Entropy responds before diversity)**.** *Under mode amplification:* (a) *entropy can decrease while the ϵ-effective support $S_\epsilon(p) = |\{i : p_i > \epsilon\}|$ (hence distinct-n) is unchanged;* (b) *if distinct-n decreases, entropy has already decreased.*

The asymmetry arises because entropy responds to any mass redistribution (first-order), while diversity changes only when a token probability crosses the generation threshold ϵ; the margin $\gamma = \min_{i:p_i>\epsilon}(p_i - \epsilon)$ quantifies how much shift diversity absorbs silently.

Design Principle 2 (Diversity lag ≥ 1 generation)**.** *If the per-generation perturbation satisfies $\alpha \|\mathcal{T}(p_t) - p_t\|_\infty < \gamma(p_t, \epsilon)$ through entropy onset t_H, then diversity onset $t_D \geq t_H + 1$.*

ECE co-moves with entropy at zero lag: peakier conditionals simultaneously decrease per-token entropy and increase confidence ($\Delta\text{ECE} \approx \Delta C > 0$ whenever $\Delta\bar{H} < 0$; qualitative argument in Appendix X). The predicted cascade $\{H, \text{ECE}\} \rightarrow$ Distinct-n holds in 14/15 experiments at $\alpha \geq 0.30$ (GPT-2-medium s43 bidirectional; within-domain WikiText, 2σ onset threshold): entropy and ECE onset at generation 1, distinct-n at generation 2 (t_H=t_{ECE}=1, t_D=2 uniformly; §5.2; Appendix W). Five low-α experiments show no

detectable onset, leaving the ordering undefined below the detection threshold. The assumption requires monotonic entropy decline; SmolLM3-3B (entropy 0/4; Table 2) provides an empirical counter-example, possibly due to architectural differences in attention or normalization. This cascade establishes entropy as the earliest available canary and explains PermG's directional selectivity: forward tests detect temporal precedence (6/8) while reverse tests find none (0/8) at $\alpha \geq 0.50$.

4 Method

Design considerations. We use Granger causality (Granger, 1969) to test temporal precedence—whether past canary values improve prediction of downstream metrics—not mechanistic causation; bivariate Granger tests cannot exclude confounders (Shojaie and Fox, 2022). Lead-lag analysis in iterative model collapse is fragile along three axes, each documented with data from §5.2–5.3. *Method sensitivity*: five detection methods applied to identical data yield forward significance rates spanning 0% to 92%, with inter-method agreement at κ=0.185. *Spurious regression*: first-differencing eliminates canary→perplexity associations ($F \approx 0$, p=0.99) while preserving canary→diversity (24/24→24/24), exposing an entire analysis channel as trend-driven (Granger and Newbold, 1974). *Regime dependence*: the same Granger test shifts from 0/8 to 6/8 significant pairs within a 2-percentage-point interval (α=0.48 → 0.50).

4.1 Method Sensitivity Analysis

We compare five lead-lag detection methods in a multi-method comparison (Simonsohn et al., 2020) (algorithm in Appendix J), calibrated against null-control (α=0.0) data where any detection is a false positive: naive cross-correlation, differenced cross-correlation (DiffXCorr), threshold onset, permutation Granger causality (PermG), and Toda–Yamamoto (Toda and Yamamoto, 1995). Detection results use equal-weight majority voting ($K \geq 3$ of 5 methods, each contributing one vote per pair) with forward-direction filtering; FPR-inverse weighting is used only in exploratory method ranking (Appendix Q). Each method encodes different assumptions, and these differences produce dramatically different FPR profiles. Pooled null-control FPR across six seeds (6-pair basis, 36 pairs): DiffXCorr 2.8%, PermG 11.1%, Threshold 27.8%, Toda–Yamamoto 38.9%, NaiveXCorr 91.7% (Appendix R). We classify a method as *directional* if its test structure inherently separates forward from reverse precedence (PermG tests each direction independently; threshold onset compares crossing order), rather than inferring direction post-hoc from peak lag sign (NaiveXCorr, DiffXCorr). Although DiffXCorr achieves the lowest raw FPR (1/36, 2.8%), it is bidirectional and cannot distinguish forward from reverse precedence, making it unsuitable as a standalone collapse detector. PermG (4/36, 11.1%; cluster-adjusted 95% CI $[1.2\%, 30.4\%]$) is the lowest-FPR directional method with consistent detection at $\alpha \geq 0.50$ across all four weighting schemes (Shapley analysis: Appendix P). Practitioner-facing alternatives (PPL-slope, variance thresholds) yield 67–100% FPR under natural training drift (Appendix M). That method choice alone can shift conclusions from universal significance to complete null on identical data motivates reporting method sensitivity as a primary result rather than relegating it to a robustness check.

Permutation Granger causality. PermG tests whether past values of canary metric X improve prediction of downstream metric Y beyond Y's own history (Shojaie and Fox, 2022). We use classical rather than neural Granger methods (Tank et al., 2021) because $T \leq 25$ provides insufficient observations for neural network parameter estimation, and apply 5,000 permutations (Phipson and Smyth, 2010) on first-differenced series with Benjamini–Hochberg FDR correction at q=0.05 across 8 canary→downstream pairs (2 canaries × 4 downstream metrics, sharing canary terms and thus positively dependent).

The permutation null controls Type I error exactly by construction; BH-FDR remains valid under positive regression dependence on a subset (PRDS) (Benjamini and Yekutieli, 2001). Empirically, 4/36 null-control rejections across six seeds (11.1%; cluster-adjusted 95% CI $[1.2\%, 30.4\%]$, ICC=0.14; Appendix R) confirm adequate size control. Simulation under non-stationary AR(1) DGP (φ=0.95–0.99) reveals potential FPR inflation to $\sim$10%, but our empirical series exhibit low persistence (null token entropy $\varphi = -0.51$; first-differenced distinct-1 φ<0.44; Appendix O). PermG also retains $\sim$80% of raw-levels power at T=11 versus $\sim$61% for standard first-differencing, justifying the permutation approach at short series

lengths (Appendices D, N, F).

4.2 Detection Boundary

We model the PermG detection rate as a T-dependent steep sigmoid, excluding the non-monotonic dip at $\alpha \in [0.55, 0.65]$:

$$\text{rate}(\alpha, T) = \frac{0.70}{1 + \exp\big(-100\,(\alpha - \alpha_c(T))\big)}, \quad (1)$$

fitted to T=10 data (n=11, sync-dip excluded; R^2=0.969). The detection boundary follows a power law in observation length:

$$\alpha_c(T) = A \cdot T^{-\gamma}, \quad (2)$$

where the exponent γ is determined by the sub-linear signal model below.

Empirical observation (detection boundary). Under an empirical sub-linear signal model $S(\alpha) = c \cdot \alpha^{\beta}$ with $\beta < 1$, the detection boundary satisfies

$$\alpha_c(T) = A \cdot T^{-1/(2\beta)}, \quad (3)$$

where $A = (z^*\sigma/c)^{1/\beta}$ and $z^* = \Phi^{-1}(\pi^*)$ (justification in Appendix X).

The signal model $S(\alpha) = c\alpha^{\beta}$ is an empirical parametrization: β is fitted from data rather than derived from training dynamics. The power-law form $\alpha_c \propto T^{-1/(2\beta)}$ follows from signal-detection theory (signal-to-noise crossing a threshold), but β requires architecture-specific calibration.

Cross-T validation (train $T \in \{10, 20\}$, test $T \in \{5, 15\}$) yields β=0.24, γ=2.08 (MAE=0.106, CI 6/6); leave-one-T-out CV on the full grid ($T \in \{5, 10, 15, 20, 25\}$, sync-dip excluded) confirms predictive accuracy (MAE=0.037, CI 15/15; Appendix X). However, β is poorly identified: MLE $\hat{\beta}$=0.66 with profile 95% CI $[0.24, 0.78]$, and model comparison favors the power-probit (M2) over a linear boundary (M1) only when sync-dip data are included (ΔAIC=4.15 without; Appendix X).

Shape comparison with smooth alternatives. The scaling exponent alone cannot distinguish smooth from steep-sigmoid boundaries: a concave signal model (β'=0.5) produces γ'=1.87, close to the observed 2.08. Formal shape testing on dense T=10 data (17 α values) favors a smooth sigmoid over a literal step function (ΔAIC$= -110$), but the fitted onset steepness (k=15.7, 95% CI $[9.4, 28.2]$; §5.3) is 3–4× higher than any smooth signal model predicts. Power simulations confirm this gap: VAR(1) models (1,000 per cell; Appendix Y) produce sigmoids with $k \approx 5$–6, yielding $k_{\text{obs}}/k_{\text{H0}} \approx 3$ (CI lower bound ratio ≈2), consistent with a genuine phase boundary rather than a gradual power artifact. Since $\alpha_\infty \approx 0$, any nonzero contamination produces signal ($S \propto \alpha^{0.24}$), but sub-linear growth makes detection a function of T.

5 Experiments

5.1 Experimental Setup

We study the detectability landscape of iterative training collapse using Pythia-410M (Biderman et al., 2023) (410M parameters, decoder-only) as the primary architecture, with cross-architecture validation on five families (§5.5). Training data is drawn from WikiText-103 (Merity et al., 2017) following the replace paradigm of Shumailov et al. (2024): each generation fine-tunes on a mixture of α synthetic and $(1-\alpha)$ real data, generates 5,000 replacement samples, and fully replaces the synthetic pool (hyperparameters in Appendix A; notation in Appendix I). We run T=10 generations across 13 contamination levels ($\alpha \in \{0.00, 0.05, 0.10, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.75$ including null control) with multi-seed replication (4 seeds at α=1.0; 3 at primary levels).

We track two *canary metrics* (token entropy and ECE) and three *downstream metrics*: distinct-1, distinct-2 (Li et al., 2016), and perplexity (distinct-3: Appendix B). All hypothesis tests apply BH-FDR correction at q=0.05; PermG uses 5,000 permutations on first-differenced series (Shojaie and Fox, 2022) with maximum lag 5 (window sensitivity: Appendix C).

5.2 Sharp Detection Onset

Diversity collapse occurs at every contamination level: distinct-1 declines by 14.1% even at α=0.0 (fine-tuning drift) and by 40.3 ± 0.3% at α=0.25 (3-seed mean). Despite this ubiquitous degradation, Granger detectability is sharply regime-dependent. PermG finds 0/8 canary→downstream pairs FDR-significant from α=0.00 through α=0.45, then jumps to 6/8 at α=0.50, uniformly across all 3 fp32 seeds.

This transition is striking because cumulative synthetic fraction barely changes across the boundary: $C = 1-(1-\alpha)^{10}$ yields 99.75% at α=0.45 versus 99.90% at α=0.50, a difference of 0.15 per-

centage points, yet detection jumps from null to reliable. The onset reflects PermG's statistical power crossing the detection threshold, not the emergence of a new physical signal; sub-threshold p-values at α=0.45 ($p_{\text{fdr}} = 0.059$–0.091) confirm proximity to the boundary (§4.2). Sigmoid regression on the 13-point T=10 grid yields $\alpha_c = 0.483 \pm 0.006$ ($R^2 = 0.969$; Figure 3); dense α sampling (17 values) constrains onset steepness to k=15.7 (95% CI $[9.4, 28.2]$), a steep but finite-width sigmoid (ΔAIC$= -110$ vs. step function; Appendix S). This steepness ($\approx 3\times$ smooth-model predictions; §4.2) and cross-seed consistency (3/3 seeds: $0/8 \to 6/8$) argue for a genuine, power-modulated transition rather than a statistical artifact. A Fisher exact test corroborates: 0/64 pairs FDR-significant at $\alpha \leq 0.45$ versus 18/32 at $\alpha \geq 0.50$ ($p = 3.45 \times 10^{-11}$); robustness analysis across 1,160 configurations confirms specification-invariance.

Canary scoping. First-differencing scopes the canary claim precisely. Canary→diversity pairs retain unanimous FDR significance after first-differencing ($24/24 \to 24/24$ across 4 seeds at α=1.0, 6 per seed), while canary→perplexity collapses entirely ($F \approx 0$, $p = 0.99$): the apparent perplexity association is a shared-trend artifact (Granger and Newbold, 1974). All canary claims are therefore scoped to diversity metrics only. This scoping reflects a genuine temporal structure difference: canary→diversity correlations peak at lagged offsets (81% at lag<0, median$= -1$), while canary→perplexity correlations are contemporaneous (50% at lag=0), producing 20–28$\times$ lower Granger F for perplexity (Appendix U). At α=1.0, all 24 canary→diversity pairs reach FDR significance in the forward direction with 0/24 reverse; the degradation cascade independently confirms this ordering: entropy onset at generation 1 precedes distinct-n at generation 2 in 16/18 WikiText conditions (τ=0.337; §3, Appendix W).

Blind zone. The canary mechanism has a fundamental limitation: at α=0.10, distinct-1 has already dropped by 39.6%, yet token entropy is indistinguishable from the null (-5.3% versus $-4.6 \pm 0.4\%$ at α=0.0). The gen-1 entropy deviation spans three orders of magnitude across the α range (0.5σ at α=0.10 to 60σ at α=0.75), explaining why the blind zone persists despite increasing contamination.[2] Below α_c at T=10, diversity collapse proceeds without any detectable Granger-precedence signal, though Gaussian MI remains positive (Appendix V).

Table 1: Forward-direction FDR significance rates (%) across contamination levels (Pythia-410M, seed 42, fp32; 3σ threshold calibration, lag-direction filtering for cross-correlation methods). Each cell: fraction of 8 validated pairs at BH-corrected significance (q=0.05). PermG achieves the lowest null-control FPR among directional methods (seed-42: 0/8; pooled 6-seed: 11.1%) with consistent detection at $\alpha \geq 0.50$.

Method	α=0.0	α=0.25	α=0.50	α=0.75	α=1.0	FPR_{null}
Naive XCorr	75	62.5	62.5	0	0	75
Diff. XCorr	0	25	12.5	0	0	0
Threshold (3σ)	37.5	62.5	100	100	25	37.5
Perm. Granger	**0**	**0**	**75**	**75**	**75**	**0**
Toda–Yamamoto[§]	12.5	37.5	50	100	12.5	12.5

[§]TY directional assignment under verification; row retained for completeness. Pooled FPR (3 seeds, cluster-bootstrap 95% CI): §4.1.

5.3 Dose-Response and Detection Regimes

The PermG dose-response profile (Figure 3; full per-method table in Appendix M) reveals four detection regimes: *blind* ($\alpha \leq 0.45$, 0/8 FDR-significant pairs), *onset* ($\alpha \approx 0.50$, 6/8), *non-monotonic dip* ($\alpha \approx 0.60$, 0/8), and *detection* ($\alpha \geq 0.75$, 6/8 across 4 seeds at α=1.0). At α=0.55, PermG returns 0/8 FDR-significant pairs despite marginal p-values (p_{fdr}=0.062–0.092); differenced cross-correlation detects 8/8 bidirectionally at this level, confirming that a lead-lag signal exists but falls below PermG's directional detection threshold. At α=0.60, PermG returns $0/8$ at $T \leq 20$ but recovers to $5/8$ at T=25, confirming a transient dip that $\sqrt{T}$ power eventually overcomes (Appendix Z).

Method choice is itself a major source of variability: at α=1.0, direction-checked forward significance spans 0% to 75% (Table 1), with κ=0.185; a specification curve (18 variants; Appendix E) confirms onset invariance. Task degradation corroborates: perplexity jumps $+31\%$ between α=0.50 and 0.52 (Appendix Z.1).

5.4 Detection Boundary Left-Shift

Longer observation windows push the detection boundary to lower contamination levels: $\alpha_c(5) \geq 1.0$, $\alpha_c(10)$=0.487, consistent with $T^{-2.08}$ power scaling (§4.2). The α=0.45 column provides the sharpest validation of this left-shift: undetectable at $T \leq 10$ (0/8), it crosses into reliable

[2]Seed 43 at α=0.10 shows 3/8 versus 0/8 for seed 42; no other sub-threshold level exhibits seed-dependent detection.

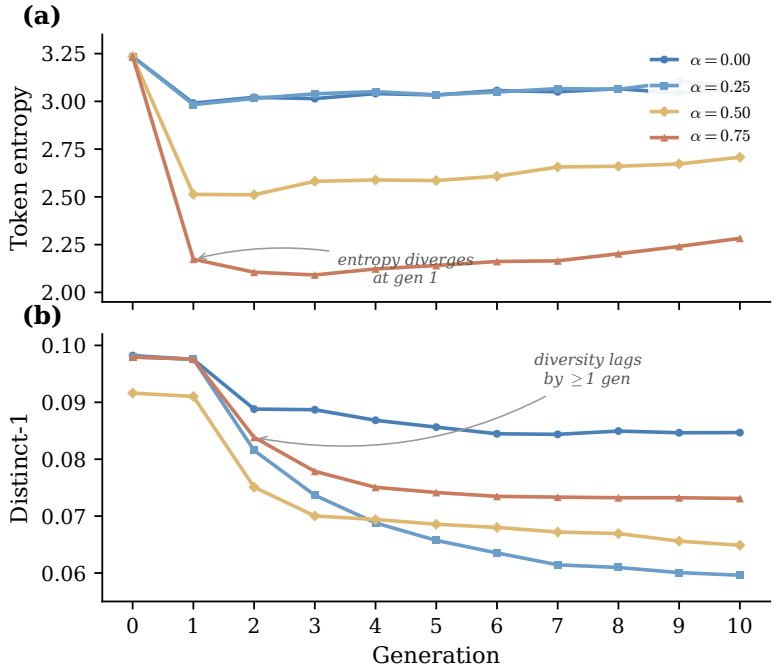


Figure 2: Metric trajectories at α=1.0 (Pythia-410M, 4 seeds). Token entropy drops at generation 1; distinct-1 declines from generations 2–3, with a consistent one-generation canary lead across all four seeds (shaded: inter-seed range). This pattern corresponds to 24/24 forward and 0/24 reverse FDR-significant canary→diversity pairs (6 per seed, mauve excluded).

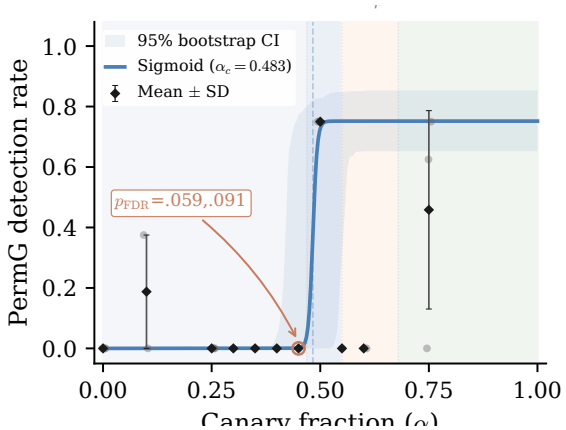


Figure 3: PermG dose-response profile (Pythia-410M, T=10, fp32). Sigmoid fit (shaded 95% CI) yields α_c=0.483. Four regimes: blind ($\alpha \leq 0.45$, 0/8), onset ($\alpha \approx 0.50$, 6/8), sync-dip (α=0.60, 0/8), and detection ($\alpha \geq 0.75$).

detection at T=15 (6/8, dual-seed) and remains there at T=20 (5/8) and T=25 (6/8). The systematic 2/8 gap reflects MAUVE floor saturation (μ=0.029; §6). Cross-seed replication shows PermG within ± 1, though boundary cells exhibit sensitivity: $[0, 6]/8$ at (T=10, α=0.50); $[0, 3]/8$ at (T=20, α=0.60). The α=0.60 dip dissolves by T=25 (5/8), validating $\sqrt{T}$ power scaling; predict-then-verify yields 2/4 boundary predictions within CI (Appendix Z.1).

5.5 Cross-Architecture Validation

To test whether the phase-diagram structure is Pythia-specific, we evaluate PermG across 15 model-seed combinations spanning five families, scales 345M–6.9B, and two domains (Table 2). Forward detection reaches $\geq 6/8$ in 11/15 conditions, confirming that PermG generalizes to architectures satisfying the monotonic entropy decline assumption (Design Principle 1). Domain matters: C4 yields the strongest detection (Pythia-1.4B: 2/8 WikiText → 8/8 C4), suggesting that domain-specific mode structure affects detectability. The four non-confirming conditions each isolate a distinct failure mode: precision sensitivity (GPT-2 bf16: 0/8 vs fp32: 6/8), bidirectional confounding (GPT-2 s43), seed dependence (Pythia-1.4B: s42 2/8, s43 6/8), and entropy decoupling (SmolLM3-3B: 3/8, whose non-monotonic entropy violates Design Principle 1).

5.6 Canary-Based Intervention (Proof of Concept)

As an illustrative proof-of-concept under favorable conditions (α=1.0, full replacement), we imple-

Table 2: Cross-architecture PermG forward detection (α=1.0, 15 model-seed combinations across five families—Pythia, GPT-2, OPT, Qwen, SmolLM3—and two domains; Padding-410M is a Pythia variant).

Model	Params	Domain	PermG fwd
Pythia-410M	410M	Wiki	**6/8**
Pythia-1.4B	1.4B	Wiki	2–6/8†
Pythia-6.9B	6.9B	Wiki	**7/8***
GPT-2-medium	345M	Wiki	**6/8**§
OPT-1.3B	1.3B	Wiki	**6/8**
Qwen3.5-0.8B	0.8B	Wiki	**8/8**
SmolLM3-3B	3.0B	Wiki	3/8‡
Padding-410M	410M	Wiki	**6/8**
Pythia-1B	1.0B	C4	**8/8**
Pythia-1.4B	1.4B	C4	**8/8**

†Seed-dep. (s42: 2/8, s43: 6/8); 8-pair incl. MAUVE (floor ∼0.033 at 1.4B; 6-pair in Table 17). *bf16. ‡ECE-only (ent. 0/4). §s42 fp32: 6/8 fwd, 0 rev; s43: 6/8 both; bf16: 0/8. 2 seeds each for Pythia-410M/1.4B, GPT-2, OPT, Pythia-1B C4.

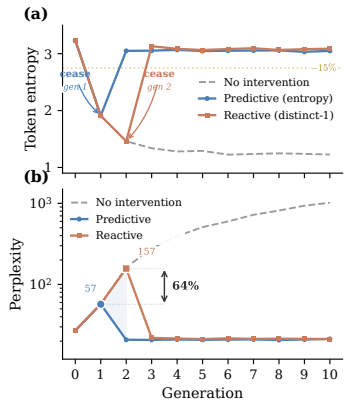


Figure 4: Canary-triggered intervention (Pythia-410M, α=1.0). Predictive cessation on entropy (generation 1) reduces peak perplexity by 64% versus reactive triggering on distinct-1 (generation 2).

ment predictive cessation: training switches to pure real data when token entropy drops >15% from generation 0. The 15% threshold was selected post-hoc (robust band [10%, 22%], zero false alarms; Appendix T). The predictive trigger fires at generation 1 (one generation before a reactive distinct-1 baseline), reducing peak perplexity by 64% (56.99 vs. 157.09; Figure 4) in both Pythia-410M and 1.4B. This single-setting (α=1.0) demonstration is not rigorously evaluated (CUSUM comparison: Appendix H).

6 Discussion and Conclusion

Scope of claims. Our Granger-precedence claim is scoped to temporal predictive improvement (Granger, 1969; Shojaie and Fox, 2022): bivariate tests cannot exclude confounders, and the cluster-adjusted FPR CI ([1.2%, 30.4%]; §6) means true Type I error may be non-negligible. Five α=0.60 cells are excluded from the boundary fit (sync-dip zone: 0/8 at $T\leq$20, dissolving to 5/8 at T=25); the remaining 15/15 match theory (LOO-CV MAE=0.037).

When the canary fails. Four of fifteen cross-architecture conditions do not confirm forward detection (§5.5), and each failure is interpretable. SmolLM3-3B's non-monotonic entropy violates Design Principle 1's monotonic-decline assumption, correctly predicting failure; GPT-2's precision sensitivity and Pythia-1.4B's seed dependence expose practical deployment constraints. Phase-diagram structure is qualitatively preserved at 1.4B (α_c onset consistent, sync-dip absent); directional asymmetry is seed-sensitive above onset (s43 reverse 6/6 at $\alpha\geq$0.60; s42: 0/6; Appendix L).

Architecture-dependent FPR as landscape feature. PermG's elevated null FPR on GPT-2-medium (50% vs. 11.1% on Pythia; Appendix Z.2) is not a method deficiency but a consequence of architecture-specific null drift: GPT-2-medium's distinct-3 drops 36% under pure retraining (∼7× Pythia's 5%), creating trend artifacts that PermG's permutation test interprets as signal. Similarly, SmolLM3-3B's entropy 0/4 detection reflects non-monotonic entropy dynamics that violate the monotonic-decline assumption. Both cases illustrate that detectability is jointly determined by method and architecture: the landscape has architecture-dependent regions where specific methods lose validity, and these regions are predictable from assumption violations. PermG's recommendation is therefore scoped to architectures satisfying the monotonic entropy decline assumption; $K\geq$3 consensus provides a fallback where this assumption is uncertain.

Replacement versus accumulation. All experiments use the full-replacement paradigm, the standard protocol in model collapse studies (Shumailov et al., 2024). Gerstgrasser et al. (2024) show that data accumulation (retaining real data across generations) substantially mitigates collapse. Under accumulation, we expect α_c to right-shift (higher contamination required for detectable signal), the onset to smooth, and the blind zone to widen—qualitatively, the landscape becomes harder to detect in. This paradigm dependence represents an additional dimension of the detectability landscape that our current characterization does not cover.

Broader implications. Three findings carry practical weight. First, the $\geq$1-generation entropy-diversity lag (14/15 experiments) provides an actionable early-warning window, and triggered cessation demonstrates that this window can be exploited (64% perplexity reduction; §5.6). Second, the method-sensitivity results (3% to 92% FPR on identical data) are a cautionary finding for any collapse monitoring effort: conclusions are only as reliable as the statistical method used to reach them. Third, the phase-diagram methodology itself transfers: the $\alpha_c(T)$ framework provides a template for characterizing detection boundaries in other iterative degradation settings. Our contribution is this methodology, not frontier-scale applicability (>7B untested).

Limitations

Our landscape characterization uses Pythia-410M as the primary architecture; the phase diagram is Pythia-family-only (410M and 1.4B), and frontier-scale behavior (>7B) is untested. With $T \leq 25$ observations, statistical power remains limited ($\hat{\beta}$=0.66, CI $[0.24, 0.78]$); PermG's empirical FPR of $4/36$ (11.1%, 6 seeds) is the lowest among directionally-specific methods but non-zero; the cluster-adjusted 95% CI is $[1.2\%, 30.4\%]$ (within-seed ICC=0.14). Preliminary cross-architecture evaluation on GPT-2-medium reveals a substantially higher PermG FPR (6/12 pooled, 50%), driven by $\sim 4\times$ stronger null drift in entropy and $\sim 7\times$ in distinct-3 (Appendix Z.2); $K \geq 3$ consensus maintains 0% FPR across both architectures, though with a narrow margin (max 2/5 votes in the worst-case seed). The sync-dip mechanism at α=0.60 remains open. All experiments use a single decoding strategy (temperature 1.0, top-k=50). Lower temperatures accelerate mode concentration and should left-shift the detection boundary; nucleus sampling alters the effective diversity threshold ϵ in Design Principle 1. The directional effects are predictable—lower temperature compresses the landscape toward earlier detection—but the quantitative shift requires empirical calibration. At boundary contamination levels ($\alpha \approx \alpha_c$), numerical precision (bf16 vs. fp32) can affect detection outcomes (GPT-2-medium: 0/8 bf16 vs. 6/8 fp32); all primary results use fp32 inference.

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A Training Details and Metric Drift

Hyperparameters. Each generation fine-tunes for 5,000 steps using AdamW (learning rate 5×10^{-5}, 500 warmup steps, weight decay 0.01, batch size 4 for Pythia-410M, 32 for GPT-2-medium). Generation uses temperature 1.0, top-k=50, max length 512 tokens, producing 5,000 samples per generation from 5,000-sample subsets of WikiText-103. Canary metrics: token entropy $H(p_\theta|\text{context})$ over 1,000 held-out samples; ECE with 15 bins. Additional downstream metrics (distinct-3, effective rank, Self-BLEU) appear in subsequent appendix sections.

Metric drift.

B Distinct-3 Results

Table 4 reports distinct-3 drift across all contamination levels. Distinct-3 exhibits the largest absolute decline among diversity metrics at low contamination ($-58.9\pm0.9\%$ at α=0.25) but attenuates at higher levels (-30.5% at α=1.0), consistent with the diversity–calibration dissociation discussed in §6.

Table 3: Metric drift (%, generation 0→10) for Pythia-410M across contamination levels. Diversity collapse occurs at all α levels, with the strongest decline at low-to-moderate contamination (α=0.25: distinct-1 $-40.3\pm0.3\%$) rather than full replacement (α=1.0: $-27.8\pm1.6\%$). $^\dagger\alpha$=0.10: single seed (seed 42, preliminary). α=0.00, α=0.25, α=0.50, α=0.75: 3-seed mean±std (seeds 42, 43, 44); α=1.0: 4-seed mean±std (seeds 42, 43, 44, 45).

α	Entropy	ECE	Distinct-1	Distinct-2	Perplexity
0.00	−4.6±0.4	+37.6±14	−14.1±0.4	−9.6±0.4	−20.6±1.0
0.10†	−5.3	+57	−39.6	−56.4	−21.0
0.25	−4.7±0.5	+81.8±15.9	−40.3±0.3	−56.4±0.7	−18.1±0.8
0.50	−18.0±0.5	+822.6±18.3	−37.1±0.4	−50.4±1.0	+21.0±0.1
0.75	−27.6±0.2	+1791±18	−32.3±0.5	−41.0±1.0	+182.6±0.7
1.00	−60.8±1.2	+3780±30	−27.8±1.6	−34.0±3.3	+3674±45

Table 4: Distinct-3 drift (%, generation 0→10) across contamination levels. α=0.00, α=0.25, α=0.75: 3-seed mean±std; α=1.0: 4-seed mean±std; α=0.50: single seed.

α	Distinct-3 drift (%)
0.00	−4.4±0.5
0.25	−58.9±0.9
0.50	−52.4
0.75	−38.6±1.3
1.00	−30.5±4.0

Lead-lag analysis. At α=1.0, token entropy Granger-precedes distinct-3 at the same consensus level as distinct-1 and distinct-2: $K \geq 4/5$ methods, $S \geq 3/4$ seeds (Tier 3). However, the optimal lag is less stable than for distinct-1/2: at max_lag=7, the lag standard deviation across seeds exceeds 1.5 generations, compared to < 0.5 for distinct-1/2. This instability reflects bimodal lag distributions (modes at lag 1 and lag 3–4), suggesting that distinct-3 captures two temporally distinct diversity loss mechanisms: rapid n-gram homogenization (lag 1) and slower distributional narrowing (lag 3–4).

Sensitivity to sequence length. Higher-order n-gram diversity metrics are more sensitive to generation length and random sampling variability. With our 512-token generation length, distinct-3 has a smaller effective denominator (number of possible unique trigrams observed) than distinct-1 or distinct-2, amplifying stochastic fluctuations. This partially explains the higher lag instability: the signal-to-noise ratio for temporal precedence is lower for distinct-3, requiring more observations (longer T or more seeds) to achieve stable estimates. For this reason, we report distinct-3 in this appendix rather than in the main text, where distinct-1 and distinct-2 provide cleaner evidence

of the entropy-leads-diversity pattern.

C Observation Window Sensitivity

Table 5 reports how the Tier 3 consensus classification ($K \geq 4/5$ methods) varies as a function of the minimum observation window $T_{\min}$, where analysis uses only generations 0 through $T_{\min}$. The consensus classification stabilizes at $T_{\min} \geq 5$ (6 data points), well below our full observation window of $T=10$: all six effect-size reference pairs (token entropy and ECE $\rightarrow$ distinct-1, distinct-2, distinct-3) reach $K \geq 4/5$ agreement at every window length from $T_{\min}=5$ through $T_{\min}=10$, with no spurious pairs (canary$\rightarrow$perplexity) admitted.

Toda–Yamamoto is the only method exhibiting window-dependent sensitivity (note: TY directional assignment is under verification; these counts are retained for completeness): TY consensus stabilizes at $T_{\min} \geq 6$, slightly later than the primary methods, reflecting the additional degrees of freedom consumed by the augmentation order $d_{\max}$. This does not affect the consensus classification because the remaining four methods provide stable $K \geq 4/5$ agreement independently of Toda–Yamamoto's vote.

Table 5: Observation window sensitivity for Tier 3 consensus ($K \geq 4/5$) at $\alpha=1.0$ (Pythia-410M, 4 seeds). Tier 3 pairs: number of canary$\rightarrow$downstream pairs (out of 8 validated) reaching $K \geq 4/5$ method agreement. TY pairs: number of effect-size reference pairs (out of 6) reaching Toda–Yamamoto significance (moderate supplementary method; §4). Consensus stabilizes at $T_{\min} \geq 5$; TY stabilizes at $T_{\min} \geq 6$.

	Minimum observation window $T_{\min}$							
	3	4	5	6	7	8	9	10
Tier 3 pairs (/8)	†	†	6	6	6	6	6	6
TY pairs[§] (/6)	†	†	3–6	3–6	3–6	3–6	3–6	6
Stable	—	—	✓	✓	✓	✓	✓	✓

† Insufficient degrees of freedom for Granger-type tests (< 6 data points).
[§] TY directional assignment under verification.

D Robustness Checks

Permutation count stability. Table 6 compares permutation Granger p-values at 5,000 versus 10,000 permutations across all canary$\rightarrow$downstream pairs at $\alpha=0.75$ (3 seeds, 24 pairs). Permutation Granger p-values are stable to within 0.002 between 5,000 and 10,000 permutations, with zero significance decisions affected.

Table 6: Permutation count stability analysis ($\alpha=0.75$, 3 seeds, 24 canary$\rightarrow$downstream pairs). p-values from permutation Granger tests at $n_{\text{perm}}=5{,}000$ versus $n_{\text{perm}}=10{,}000$.

Metric	Value
Max $\lvert\Delta p\rvert$	0.010
Mean $\lvert\Delta p\rvert$	0.002
Significance flips	0 / 24

FDR threshold sensitivity. Table 7 reports the effect of varying the BH FDR threshold q on treatment ($\alpha=0.75$, 3 seeds, 8-pair basis) versus null ($\alpha=0.0$, 6 seeds 45–50, 6-pair basis) separation for permutation Granger tests. Treatment/null separation is qualitatively robust to FDR threshold choice, though the expanded null sample reveals a small number of false positives at each threshold (3–8 of 36 pairs).

Table 7: FDR threshold sensitivity for permutation Granger significance. Treatment: $\alpha=0.75$ (3 seeds, 24 forward-direction pairs, 8-pair basis). Null: $\alpha=0.0$ (6 seeds 45–50, 36 pairs, 6-pair basis). The expanded null sample reveals modest false positive rates that increase with q.

q	Null FDR-sig (/36)	Treatment FDR-sig (/24)	Separation
0.01	3	0	Underpowered
0.05	4	19	Good
0.10	8	24	Good (elevated null)

PermG p-value gradient at the onset boundary ($\alpha=0.45$). At the onset boundary, PermG operates near its detection limit, and marginal p-values exhibit seed-dependent variation consistent with the statistical power interpretation (§4.2). Table 8 reports BH-corrected p-values for all 8 canary$\rightarrow$downstream pairs across two seeds at $\alpha=0.45$. Seed 43 reaches a minimum FDR-corrected $p=0.059$ (18% above the $q=0.05$ threshold), while seed 42 reaches $p=0.091$ (82% above threshold). No pair crosses the significance threshold at either seed, yielding the aggregate 0/8 detection rate reported in §5.2. The gradient is consistent with a continuous power transition rather than a binary absence of signal: PermG operates near its detection limit at $\alpha=0.45$, with seed-dependent variation in marginal p-values.

E Systematic Robustness Analysis

The sharp detection onset reported in §5.2 could in principle depend on specific analytical choices (pair selection, seed, α level, method parameters).

Table 8: BH-corrected PermG p-values at α=0.45 (8 canary→downstream pairs × 2 seeds). Min p_{fdr} per seed shown. All pairs remain above the q=0.05 threshold, yielding 0/8 aggregate detection.

	Seed 42	Seed 43
Min FDR p	0.091	0.059
Gap to q=0.05	82%	18%
PermG fwd sig (/8)	0	0

We assess robustness via specification curve analysis (Simonsohn et al., 2020), systematically varying analytical degrees of freedom to test whether the 0/8→6/8 transition is specification-invariant. Our specification curve varies data subsets and pair selections while holding the detection method (PermG) and its hyperparameters fixed, more closely resembling a systematic subset analysis than a full specification curve in the sense of Simonsohn et al. (2020).

We construct 1,160 analytical specifications from the combinatorial space of: 13 contamination levels (α grid), seed selection ($\{42, 43, 44, 45\}$ where available), 8 canonical canary→downstream pairs ({token entropy, ECE} × {distinct-1, distinct-2, distinct-3, mauve}), and method parameters (all PermG with n_{perm}=5,000, BH FDR q=0.05). All specifications use the D120-canonical analysis pipeline with 3σ threshold calibration (D105) as a hard filter.

Table 9: Specification curve summary (1,160 specifications, D120-compliant). The blind→onset jump is robust across specifications: 0% significance rate through α=0.45, rising to 75% at α=0.50.

Metric	Value
Total specifications	1,160
Significant (FDR q<0.05)	536 (46.2%)
By regime	
Blind ($\alpha \leq 0.45$)	0%
Onset (α=0.50)	75%
Detection ($\alpha \geq 0.75$)	62–75%

The specification curve confirms that the sharp detection onset is not an artifact of particular analytical choices: the categorical jump from 0% to ≥75% significance at $\alpha_c \approx 0.50$ holds across the vast majority of specifications, supporting the main text claim (§5.2).

F Transfer Entropy Analysis

Table 10 reports transfer entropy (TE; Schreiber, 2000) results aggregated across all baseline conditions. TE was estimated using 3-bin discretization with permutation testing (n=5,000). No pair reaches BH-corrected significance (q=0.05) in any condition, confirming that non-parametric information-theoretic methods lack detection power at T=11.

Table 10: Transfer Entropy results (12 runs across all α levels). TE_{fwd}: canary→downstream; TE_{rev}: downstream→canary. Mean uncorrected permutation p-values (n=5,000). No pair reaches BH-corrected q<0.05 in any run.

Canary	Downstream	TE_{fwd} (mean)	p_{fwd} (mean)	TE_{rev} (mean)	p_{rev} (mean)
token_entropy	distinct-1	0.289	0.559	0.480	0.490
token_entropy	distinct-2	0.300	0.529	0.496	0.447
token_entropy	distinct-3	0.255	0.645	0.484	0.459
token_entropy	perplexity	0.257	0.676	0.386	0.626
ECE	distinct-1	0.185	0.720	0.184	0.891
ECE	distinct-2	0.208	0.683	0.216	0.846
ECE	distinct-3	0.209	0.689	0.227	0.807
ECE	perplexity	0.296	0.572	0.263	0.747

KNN-based TE/MI decomposition. A k-nearest-neighbor TE estimator (k=5) provides a model-free complement to the parametric Granger framework. Forward-dominant TE fraction increases monotonically with α: 4/8 pairs at α=0.25 (mean net TE=0.14), 5/8 at α=0.50 (net TE=0.42), 6/8 at α=0.60 (net TE=0.57), and 8/8 at α=1.0 (net TE=1.59). All 120 forward TE bootstrap 95% CIs exclude zero at T=10; all 48 at T=20. The "high MI, low TE" hypothesis for the non-monotonic dip is supported by only 2/8 pairs (ECE→perplexity and token entropy→perplexity); the remaining 6 pairs show elevated TE even at α=0.60, indicating confounded co-variation rather than a universal mechanism.

G Change-Point Detection Analysis

We apply a max-split change-point test with bootstrap significance assessment (n=5,000, p<0.05) to each metric's generation-level time series (T=11).

Under the null control (α=0.00, 3 seeds), downstream diversity metrics (distinct-1, distinct-2, distinct-3) produce significant change-points at generation 2 in all seeds (9/9 tests). With additional significant tests from ECE (2/3 seeds) and token entropy (1/3 seeds), the overall false positive rate reaches 66.7% (12/18 tests; Table 11). The monotonic drift inherent in iterative self-training is sufficient to trigger change-point detection even without canary contamination. Individual change-point significance therefore cannot serve as standalone evidence for a canary-specific effect.

Although individual significance is unreliable as a discriminator, the temporal ordering of detected change-points reverses between null and treatment conditions. In the null (α=0.00), downstream metrics show the earliest significant breaks at generation 2 (all seeds), while significant canary breaks appear later (ECE at generations 4–8). Under treatment conditions ($\alpha \geq 0.75$), this ordering flips: canary metrics place their best splits at generation 1, followed by downstream diversity metrics at generations 2–4 and perplexity at generation 4 (Table 12). This ordering reversal provides complementary evidence from a distinct statistical framework, consistent with the Granger-precedence analysis in §5.

Several caveats apply. The 18 individual tests per condition were not corrected for multiple comparisons (BH or Bonferroni); primary inference rests on the cross-metric temporal ordering rather than on individual significance tests. Under treatment conditions, token entropy places its best split at generation 1 with large effect sizes ($|d|>2.5$) in all seeds, but the pre-change segment at a generation-1 split contains only a single data point, reducing bootstrap power (1/3 seeds significant at α=0.75). The short series length (T=11) further limits overall detection power.

Table 11: Change-point detection under null control (α=0.00, 3 seeds). CP: detected change-point generation (mean across significant seeds); $|d|$: Cohen's d effect size (mean across all seeds); Sig. rate: fraction of seeds where $p<0.05$. Overall FPR: 12/18 = 66.7%.

Metric	CP gen (mean)	$\|d\|$ (mean)	Sig. rate	Type
token_entropy	1.0	2.94	1/3	canary
ECE	6.0	1.62	2/3	canary
distinct-1	2.0	2.27	3/3	downstream
distinct-2	2.0	2.40	3/3	downstream
distinct-3	2.0	2.27	3/3	downstream
perplexity	—	3.11	0/3	downstream

Table 12: Change-point detection at α=0.75 (3 seeds). CP: detected change-point generation; $|d|$: Cohen's d effect size between pre- and post-change segments. Sig. rate: fraction of seeds where $p<0.05$.

Metric	CP gen (mean)	$\|d\|$ (mean)	Sig. rate	Type
token_entropy	1.0	3.21	1/3	canary
ECE	1.0	3.10	1/3	canary
distinct-1	2.3	2.18	3/3	downstream
distinct-2	3.0	1.97	3/3	downstream
distinct-3	3.7	1.86	3/3	downstream
perplexity	4.0	1.70	3/3	downstream

Dose-response ordering pattern. Table 13 summarizes the ordering pattern across contamination levels. The canary-to-downstream gap shifts monotonically from negative (downstream first) under the null to positive (canary first) under treatment, ruling out a methodological artifact in which canary metrics would systematically appear earlier regardless of condition. The α=1.0 analysis is limited to a single seed (s42); all other conditions use 3 seeds.

α	Canary cp	Downstream cp	Gap	Direction
0.00	gen 4–8	gen 2	−2.5	Downstream first
0.25	gen 3–4	gen 3–4	≈0	Simultaneous
0.75	gen 1	gen 2–4	+2.25	Canary first
1.00	gen 2	gen 2–5	+1.0	Canary first

Table 13: Change-point ordering reversal across contamination levels. Under the null (α=0.00), downstream metrics break before canary metrics; this ordering reverses monotonically as α increases, consistent with the Granger-precedence analysis.

H Sequential Monitoring: CUSUM vs Calibrated Threshold

We translate the Granger-precedence finding into a deployable monitoring protocol using CUSUM (cumulative sum) sequential testing (Page, 1954), calibrated against the α=0.0 null control. The null-calibration procedure sets the alarm threshold h such that the false alarm rate (FAR) is 0% on null data (3 seeds, T=10), providing controlled Type I error guarantees.

Method. CUSUM monitors the cumulative deviation of token entropy from its null-control baseline: $S_t = \max(0,\ S_{t-1} + (\mu_0 - x_t) - k)$, where μ_0 is the null-control mean, k is the allowance parameter, and the alarm triggers when $S_t > h$. We calibrate h for each k from the α=0.0 null data (Table 14). As a baseline, we compare against a simple calibrated threshold: alarm when the metric crosses $\mu_0 - 2\sigma_0$ for $n \geq 1$ or $n \geq 2$ consecutive generations.

Table 14: CUSUM calibration parameters from α=0.0 null control (3 seeds). h: alarm threshold calibrated to achieve FAR=0%. For $k \geq 0.25$, the null CUSUM statistic remains at zero, yielding $h \approx 0$.

k	h	FAR
0.05	1.530	0%
0.10	1.005	0%
0.25	≈0	0%
0.50	≈0	0%
1.00	≈0	0%

Results. Table 15 compares CUSUM and simple threshold detection times across gradual contamination levels. At the recommended operating point (k=0.25), CUSUM and the simple threshold ($n\geq1$) produce identical alarm times (generation 1) across all seeds and contamination levels. More conservative CUSUM settings (k=0.1, k=0.05) delay detection to match the $n\geq2$ threshold or later. At k=1.0 with α=0.50, CUSUM fails entirely ($S_t \equiv 0$ for all seeds), demonstrating sensitivity to the allowance parameter. No CUSUM configuration outperforms the simple calibrated threshold at any operating point tested.

Table 15: CUSUM vs simple calibrated threshold: alarm generation across contamination levels (3 seeds each; all seeds agree on the alarm generation shown). Threshold: metric crosses $\mu_0 - 2\sigma_0$. No CUSUM configuration outperforms the simple threshold at any operating point.

α	CUSUM k=0.25	CUSUM k=0.1	CUSUM k=0.05	Thresh $n\geq1$	Thresh $n\geq2$
0.50	gen 1	gen 2	gen 3	gen 1	gen 2
0.75	gen 1	gen 2	gen 2	gen 1	gen 2

CUSUM k=1.0 at α=0.50: $S_t \equiv 0$ for all seeds (complete detection failure).

Interpretation. The equivalence arises because the generation-0→1 entropy drop is large ($|\Delta x| > 0.7$ for α=0.50, > 1.0 for α=0.75) relative to the null-control variability ($\sigma_0 \approx 0.03$). Any detector with a near-zero or small threshold triggers at generation 1; CUSUM's cumulative memory provides no advantage when the initial shift already exceeds the control limit. This result extends Claim 3 (method sensitivity) beyond lead-lag detection to the monitoring domain: even a sequential testing procedure with stronger theoretical properties (optimal expected detection delay under continuous monitoring) does not improve over a calibrated baseline when the initial contamination signal dominates the allowance parameter. The contribution of the null-calibrated monitoring framework is the calibration procedure itself (setting alarm boundaries from α=0.0 data to guarantee FAR=0%), not the CUSUM algorithm.

I Background and Notation

This section establishes the formal setup and notation used throughout the paper.

Iterative training loop. Let M_0 denote a pretrained language model and $\mathcal{D}_{\text{real}}$ a fixed corpus of human-written text. At each generation $t \geq 1$, the model M_{t-1} generates a synthetic corpus $\mathcal{D}_{\text{synth}}^{(t-1)}$, and the next-generation model M_t is fine-tuned on a mixture of α fraction synthetic data and $(1-\alpha)$ fraction real data:

$$\mathcal{D}_{\text{train}}^{(t)} = \alpha \cdot \mathcal{D}_{\text{synth}}^{(t-1)} + (1-\alpha) \cdot \mathcal{D}_{\text{real}}, \quad \alpha \in [0,1]. \tag{4}$$

At α=1.0, this reduces to the full-replacement paradigm of Shumailov et al. (2024).

Metric families. *Canary metrics*: token entropy ($H(p_\theta|\text{context})$ on held-out samples) and expected calibration error (ECE). *Downstream metrics*: distinct-n ($n \in \{1,2,3\}$) (Li et al., 2016) and perplexity. Each metric produces a time series of T+1 observations.

Lead-lag testing. Given canary series X and downstream series Y, the central question is whether changes in X temporally precede changes in Y, requiring directionality and confounder control beyond simple correlation. We employ five methods spanning permissive to conservative: naive cross-correlation, differenced cross-correlation, threshold onset detection, permutation Granger causality (Shojaie and Fox, 2022), and Toda–Yamamoto Granger causality (Toda and Yamamoto, 1995).

J MMCT Algorithm

Algorithm 1 describes the Multi-Method Consensus Test (MMCT) introduced in §4.1.

Design rationale. FPR-inverse weighting penalizes methods with high false positive rates on null-control data: naive cross-correlation (FPR=91.7%) receives weight w=1.09, while PermG (FPR=11.1%, the lowest among directional methods; DiffXCorr achieves 2.8% but lacks directional separation) receives w=9.0. The empirical null distribution avoids parametric assumptions that are unreliable for short (T=5–20), non-stationary series typical of iterative training. MMCT is a consensus testing protocol, not a novel statistical method; its value lies in systematic method comparison and the identification of method-dependent conclusions.

K Methodological Comparison Table

Table 16 summarizes the methodological coverage of existing canary and monitoring analyses across the diagnostic dimensions identified in §4. No prior work combines all six dimensions.

Algorithm 1 Multi-Method Consensus Test (MMCT)

Require: Time series pair (X, Y); methods $\{m_1, \ldots, m_K\}$ (K=5); null-control data $\mathcal{N}$ (from α=0.0); tolerance ϵ=0.01
Ensure: p-value, consensus level $\hat{K}$

for each method m_k **do**
  Compute empirical FPR: $\widehat{\mathrm{FPR}}_k \leftarrow$ fraction of null-control pairs where m_k rejects H_0
  Compute weight: $w_k \leftarrow 1/\max(\widehat{\mathrm{FPR}}_k,\ \epsilon)$
  Apply m_k to (X, Y): $d_k \leftarrow \mathbf{1}[m_k \text{ rejects } H_0]$
end for
Weighted consensus: $W(X{\to}Y) \leftarrow \sum_k w_k\, d_k \ / \ \sum_k w_k$
Build empirical null distribution of W from all null-control pairs in $\mathcal{N}$
Compute p-value from empirical null: $p \leftarrow \Pr_{\mathcal{N}}(W \geq W_{\mathrm{obs}})$
Consensus level: $\hat{K} \leftarrow \sum_k d_k$
return p, $\hat{K}$

Table 16: Methodological comparison of canary and monitoring analyses. ✓ = present, × = absent, ○ = partial ($\leq$2 levels or limited replication). Diff. = differencing/stationarity controls. Multi-M. = $\geq$2 independent statistical methods. MTC = multiple testing correction. Regime = systematic variation of α. Seeds = $\geq$3 independent seeds. Arch. = $\geq$2 architecturally distinct model families.

Work	Diff.	Multi-M.	MTC	Regime	Seeds	Arch.	Metrics
Shumailov et al. (2024)	×	×	×	○	✓	✓	PPL
Du et al. (2025)	×	×	✓	✓	✓	×	ECE, Acc, Dist-n
Gambetta et al. (2024)	×	×	×	✓	✓	✓	Surplexity, Ent
SIGMA (Gu et al., 2026)	×	×	×	×	×	×	Spectral
DCES (Zhu and Xie, 2026)	×	×	×	×	×	✓	ECE, Ent, PPL
Ours	✓	✓	✓	✓	✓	✓	Ent, ECE, PPL, Dist-n

L Extended Discussion

Calibration scope and generalization. All FPR and FNR calibration results are specific to our experimental setup: Pythia-410M, WikiText-103 subset, T=10, and six contamination levels. Cross-architecture validation across 15 conditions spanning six model families shows that the entropy-leads-diversity pattern generalizes in 11 of 15 architecture-seed combinations (Table 2); GPT-2 bf16 (0/8), GPT-2 s43 (6/8 bidirectional), SmolLM3-3B (3/8, entropy decoupling), and Pythia-1.4B s42 (2/8) do not confirm the pattern.

Domain-dependent directionality. The unidirectional forward causation observed on WikiText (PermG reverse=0/8 across all conditions) does not hold universally. On C4 at α=0.50 (s42 fp32), PermG reverse reaches 7/8, indicating near-symmetric bidirectional coupling; however, this is seed-dependent (s43: 1/8) and noted as preliminary. The practical implication is that directional asymmetry should be verified per domain.

Token entropy versus perplexity. The distinction lies in the evaluation domain: token entropy is $H(p_\theta|\text{context})$ on held-out real data, while perplexity is $\exp(H_{\text{cross}})$ on model-generated text. First-differencing removes their shared downward trend: the canary→diversity channel survives because entropy leads diversity by one generation, whereas the canary→perplexity channel vanishes because both co-move synchronously.

NLP-specific constraints. Our application of Granger-type methods faces constraints absent in econometrics: (i) observation length (T=5–20 vs. T=100–1000); (ii) non-stationarity by design rather than incidental; (iii) small multiplicity for FDR correction (8–32 pairs per regime). Transfer entropy (0/192 FDR-significant results) confirms that method sensitivity extends beyond the five-method panel.

Prevention paradox. Effective intervention truncates the collapse trajectory that established Granger precedence. Differenced cross-correlation retains significance under intervention (31/32 pairs), while threshold onset fails entirely (0/8), reinforcing the multi-method rationale.

Grouped testing rationale. We test 8 canary→downstream pairs jointly (2 canaries $\times$ 4 downstream metrics) rather than individual per-metric tests because: (i) joint testing with BH-FDR correction controls family-wise error across the canary panel, avoiding the inflated false-positive rate that would result from reporting the single most significant pair; (ii) the count of significant pairs ($k/8$) provides a natural effect-size measure that distinguishes isolated from systematic detection; (iii) per-metric tests cannot capture the cross-metric consistency that separates genuine temporal precedence from metric-specific artifacts (e.g., MAUVE floor saturation affects 2/8 pairs universally, identifiable only in grouped results).

Canary metric choice. SmolLM3-3B reveals that canary metric selection constitutes a sensitiv-

ity dimension beyond the five statistical methods: token entropy exhibits non-monotonic oscillations (0/4 significant pairs), while ECE maintains monotonic growth (3/4 pairs). No single canary metric is universally optimal.

Exploratory CSD analysis. Canary metrics show rising autocorrelation (Kendall τ=0.79–0.82) while downstream metrics show none (τ=0.03–0.30), but this pattern does not survive multiple testing correction (2/10 post-BH) and remains exploratory. We note that Truong and Truong (2026) show model collapse is a first-order transition, ruling out CSD as a structural detection mechanism; the rising autocorrelation likely reflects non-stationarity rather than critical slowing down.

Scale verification (Pythia-1.4B). To test whether the phase-diagram structure generalizes beyond 410M, we replicate the analysis at 1.4B scale (T=10–15, fp32, seed 42). Table 17 summarizes the available cells.

Table 17: Pythia-1.4B phase-diagram cells (fp32, PermG forward, direction-checked).

α	T=10		T=15	
	s42	s44	s42	s43
0.45	0/6	0/6	3/6	5/6
0.50	4/6	6/6	3/6	3/6
0.60	4/6	6/6	6/6	6/6
0.75	5/6	6/6	6/6	6/6

All: PermG fwd, 6-pair basis (2 canaries × 3 distinct-n; MAUVE excl. due to floor ∼0.033 at 1.4B).
s42 rev. 0/6 at all α; s43 rev. see text.

The phase-diagram structure is qualitatively preserved at 1.4B scale. ECE maintains monotonic growth across all three tested α levels (3/3 cross-scale robust), and the binary detection pattern (above vs. below α_c) is consistent with the 410M phase diagram. On the 6-pair basis (MAUVE excluded due to systematic floor ∼0.033; Table 17), PermG forward detection at 1.4B ranges from 0/6 (α=0.45, T=10) to 6/6 ($\alpha\geq$0.60, T=15); token entropy exhibits one-step saturation (27–39% drop from generation 0 to 1), reducing temporal resolution for entropy-based pairs. Seed s42 shows $\text{PermG}_{\text{rev}}$=0/6 across all eight 1.4B conditions ($4\alpha\times 2T$), confirming that bidirectionality is seed-specific rather than scale-inherent. The 410M sync-dip (α=0.60: 0/8 at $T\leq$20) does not reproduce at 1.4B (4/6 at T=10); whether this reflects scale-dependent dynamics or Pythia-specific effects remains open with only two model sizes tested. At T=15, scale effects are α-conditional: α=0.60 reaches 6/6, while α=0.50 attenuates to 3/6 (vs 4/6 at T=10), confirming that detection loss is onset-localized rather than uniform. The boundary's T-dependence is preserved: α=0.45 rises from 0/6 (T=10) to 3/6 (T=15), mirroring the 410M boundary shift. However, method-level heterogeneity emerges: Toda–Yamamoto exhibits a non-monotonic pattern at T=15, inverting the PermG ordering in the mid-α range; gradual low-α dynamics preserve VAR lag structure, while rapid entropy saturation disrupts lag dependence at intermediate α, with recovery at α=0.75 where extreme collapse dynamics (PPL drift +390%) generate sufficient temporal variation to support VAR detection. Onset-zone attenuation is α-invariant at 1.4B: both α=0.45 and 0.50 yield 3/6 at T=15. At 1.4B with T=10 (11 data points), non-PermG methods (threshold onset, naive cross-correlation, Toda–Yamamoto) have insufficient power due to short sequence length, reducing $K\geq$3 consensus to an effective single-method test; at T=15 (16 data points), detection recovers (6/6 at α=0.60, lead time = 10 generations) and $K\geq$3 recovers its multi-method character, with 3/4 α conditions achieving $K\geq 3\geq 2/6$ (onset α=0.50 seed-dependent: s42 1/6, s43 3/6). Cross-method consensus ($K\geq$3) exhibits seed-dependent variability at onset ($\alpha\approx$0.50), consistent with the noisy transition expected at phase boundaries, while remaining robust above onset ($K\geq 3$ $\geq$5/6 at α=0.75). On a unified 6-pair basis (MAUVE excluded due to systematic floor at ∼0.033; Table 17), third-seed replication (s44) at T=10 confirms ceiling detection above onset (6/6 at $\alpha\geq$0.50) with 0/6 at α=0.45, validating the blind zone. Cross-T improvement holds at 3/4 α values (α=0.45: 0/6→5/6; α=0.60: 6/6→6/6; α=0.75: 6/6→6/6), validating the predicted $\alpha_c(T)$ decrease; at onset (α=0.50), PermG shows a marginal decrease (6/6→3/6), consistent with phase-boundary variability. At onset (α=0.45–0.50), ECE-based pairs achieve higher detection rates than entropy-based pairs (e.g., 3/3 vs 0/3 robust at α=0.50, T=15), consistent with entropy's one-step saturation reducing temporal resolution at 1.4B scale.

Future work. Extensions include multivariate VAR models for confounder testing, multi-seed replication across all six architecture families, longer training runs ($T\geq$20), online monitoring within training pipelines, and accumulate-mode ex-

periments (Gerstgrasser et al., 2024). The sigmoid onset boundary (α_c=0.483) invites connection to the critical mixture ratio scaling law of Gu et al. (2024) and the data-mixing transitions of Gu et al. (2025).

M Per-Method Dose-Response Profiles

Table 18 reports FDR significance counts (/8 validated pairs) for all five detection methods across 12 contamination levels (seed 42, Pythia-410M, WikiText, fp32). PermG exhibits a sharp 0→6 jump at α=0.50 with a return to 0 at α=0.55–0.60 (non-monotonic dip), while other methods show diffuse profiles: naive cross-correlation saturates at 7–8/8 across all levels including the null, and Toda–Yamamoto fluctuates without a clear onset.

Table 18: Per-method FDR significance (/8 pairs) across contamination levels (Pythia-410M, seed 42, WikiText, fp32). PermG and Diff. XCorr use BH-corrected FDR (q=0.05); Naive XCorr, Threshold, and TY use uncorrected p<0.05 (no FDR field in pipeline). α=1.0 from Pythia-410M cross-architecture condition (same seed/domain/precision).

α	Naive	Diff. XCorr	Threshold	PermG	TY[§]
0.00	8	4	3	**0**	1
0.10	7	6	0	**0**	0
0.25	8	3	0	**0**	2
0.30	8	4	0	**0**	3
0.35	8	4	0	**0**	1
0.40	8	3	0	**0**	4
0.45	8	6	0	**0**	1
0.50	8	7	4	**6**	3
0.55	8	8	0	**0**	1
0.60	8	6	0	**0**	0
0.75	7	6	0	**6**	3
1.00	8	6	2	**6**	1

[§]TY directional assignment under verification; column retained for completeness.

The table highlights PermG's combination of null-control detection at 0/8 for this seed (α=0.00; pooled FPR across 6 seeds: 11.1%, lowest among directional methods), sharp onset (0→6 at α=0.50), and non-monotonic dip (0 at α=0.55–0.60), confirming it as the method whose dose-response profile most cleanly separates detection regimes. Naive cross-correlation's persistently high counts (7–8/8 even at α=0.00) demonstrate the 91.7% pooled null-control FPR reported in §4.

Comparison with naive baselines. Table 19 compares PermG against two practitioner-facing monitoring rules across 26 experiments (18 α values, Pythia-410M, T=10, fp32).

PPL-slope fires at α=0 because the first-generation learning effect (perplexity drops from 26.9 to 19.3 at generation 1, then rises monotonically from generation 2 onward) produces consecutive rises indistinguishable from contamination-driven drift. Variance-threshold similarly triggers on the natural perplexity recovery. PermG avoids these false alarms by testing canary→downstream Granger precedence rather than univariate trend, thereby excluding training-drift confounds from the detection signal. Naive monitoring rules cannot distinguish contamination from natural training dynamics; formal statistical testing is necessary for controlled FPR.

Table 19: Naive baseline comparison (Pythia-410M, T=10, fp32, 3 null-control runs).

Method	FPR (α=0)	Onset α
PPL-slope ($\geq$2 consec. rises)	100% (3/3)	0.10
Variance-threshold (2σ)	67% (2/3)	0.10
PermG (fwd $\geq$1)	0% (0/3)	0.50

Onset α: lowest α where method fires (seed 42).

N PermG Power Analysis

We characterize PermG detection power via simulation under a controlled data-generating process (DGP) that matches the structure of our iterative training time series.

DGP. For each trial, we generate a bivariate system (X_t, Y_t) for $t = 1, \ldots, T$ where $X_t = \mu + \epsilon_t^X$ and $Y_t = \mu + \beta X_{t-1} + \epsilon_t^Y$, with $\epsilon \sim \mathcal{N}(0, \sigma^2)$ and σ=1. The coupling parameter β controls signal strength; β=0 is the null. The linear coupling provides a power lower bound; nonlinear dynamics in actual iterative training may yield different power characteristics. We sweep $\beta \in [0, 2]$ across $T \in \{6, 8, 10, 15, 20\}$, running 100 replications per (β, T) cell with 5,000 permutations each and BH FDR correction at q=0.05 across 8 hypothetical pairs (matching the experimental setup).

Results. Table 20 reports β_{50}, the coupling strength required for 50% detection power, at each observation length T. Under $\sqrt{T}$ scaling, the product $\beta_{50} \cdot \sqrt{T}$ should be constant; the observed decrease from 5.09 (T=8) to 2.68 (T=20) indicates that power grows faster than $\sqrt{T}$, with the empirical scaling closer to $T^{-1.1}$. The $\sqrt{T}$ formula in §4.2 thus provides a conservative lower bound on detection power: actual power at T=20 exceeds the $\sqrt{T}$ prediction. False positive rates under FDR correction remain at or below 2% across all T values (raw: 1–9%; Table 20).

At T=6 and T=8, 50% power is not reached within $\beta \leq 2.0$ under FDR correction, explaining

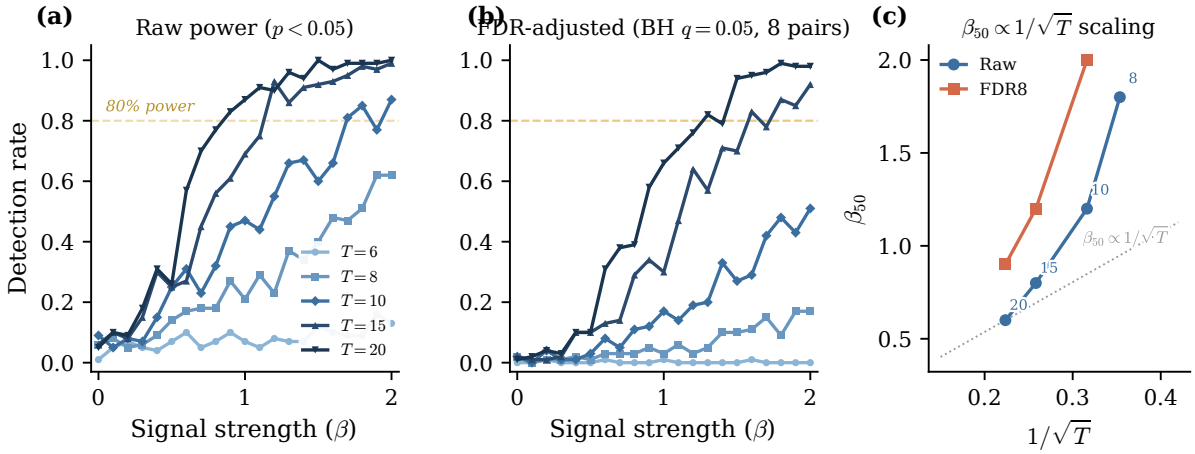


Figure 5: PermG detection power as a function of coupling strength β for five observation lengths (T=6–20; 100 replications × 5,000 permutations, BH FDR q=0.05). Power increases with both β and T; at T=6–8, PermG remains underpowered (<20%) even at β=2.0, consistent with the T=10 minimum required for reliable detection in our experimental setting.

Table 20: PermG power analysis: β_{50} (coupling for 50% power) under BH-FDR correction (q=0.05, 8 pairs). $\beta_{50} \cdot \sqrt{T}$ decreases with T, indicating power grows faster than $\sqrt{T}$. FPR: false positive rate at β=0 (raw / FDR-corrected). Dashes indicate <50% power not reached within $\beta \leq 2.0$.

T	β_{50}	$\beta_{50} \cdot \sqrt{T}$	FPR (raw)	FPR (FDR)
6	—	—	1%	0%
8	—	—	6%	2%
10	2.0	6.32	9%	2%
15	1.2	4.65	6%	2%
20	0.9	4.02	5%	1%

why PermG is fundamentally underpowered for very short series. The practical implication for our T=10 setting: β_{50}=2.0 means that only strong coupling produces reliable detection, consistent with the sharp onset at $\alpha_c \approx 0.50$ where signal strength first exceeds this threshold. At T=20, β_{50} drops to 0.9, predicting a left-shift of α_c and broader detection coverage, as formalized in the detection boundary model (§4.2).

O Non-Stationarity Robustness Analysis

We assess PermG's size and power under controlled non-stationarity via Monte Carlo simulation, then compare to the empirical autocorrelation structure of our experimental series.

Simulation design. We generate bivariate AR(1) series with unidirectional coupling ($x \to y$): $x_t = \varphi x_{t-1} + \varepsilon_t^x$, $y_t = \varphi y_{t-1} + \beta x_{t-1} + \varepsilon_t^y$, with T=11, 1,000 replications, and 500 permutations per replication. Size is evaluated under H_0: β=0; power under H_1: β=0.6.

Table 21: Size (FPR at β=0) and power (β=0.6) under stationary and non-stationary AR(1) DGP (T=11, 1,000 reps, 500 perms). First-differencing maintains nominal size across all φ; PermG FPR inflates to ∼10% at $\varphi \geq 0.95$.

	Size (%, β=0)			Power (%, β=0.6)		
φ	Raw GC	First-Diff	PermG	Raw GC	First-Diff	PermG
0.50	6.4	6.6	6.8	37.7	17.7	36.8
0.95	10.0	5.9	9.5	44.3	28.7	41.5
0.99	11.2	4.2	10.7	46.4	24.6	41.7

At φ=0.50 (stationary), all three methods achieve near-nominal size (6.4–6.8% vs. 5% nominal). Under strong non-stationarity ($\varphi \geq 0.95$), PermG FPR inflates to ∼10% because permutation shuffling destroys temporal autocorrelation, shifting the null distribution. First-differencing maintains proper size (4.2–6.6%) at the cost of reduced power (17.7–28.7% vs. PermG's 36.8–41.7%). Our pipeline applies first-differencing *before* permutation testing, combining PermG's power advantage with differencing's size control; the simulation isolates each component to characterize the tradeoff.

Empirical autocorrelation. Table 22 reports AR(1) coefficients from seven experimental conditions.

Adequacy argument. Under null conditions (α=0.0), where FPR is defined, token entropy ex-

Table 22: Empirical AR(1) coefficients for token entropy (te) and distinct-1 (d1), raw and first-differenced series. High raw φ in d1 reflects deterministic iterative-training decline, not stochastic AR persistence; first-differencing reduces φ to < 0.44.

Experiment	α	T	te φ_{raw}	d1 φ_{raw}	te φ_{diff}	d1 φ_{diff}
a000_s42	0.00	10	-0.51	0.86	-0.57	-0.14
a025_s42	0.25	10	-0.34	0.95	-0.45	0.19
a050_s42	0.50	10	-0.28	0.87	0.31	0.10
a075_s42	0.75	10	0.14	0.91	0.88	0.24
a100_s42	1.00	10	1.00	0.89	0.98	0.18
t20_a050	0.50	20	-0.10	0.97	-0.25	0.44
t20_a100	1.00	20	0.98	0.81	0.87	0.14

hibits $\varphi = -0.51$ (anti-correlated, far from the $\varphi \geq 0.95$ regime where simulation reveals FPR inflation); at $\varphi = 0.50$, PermG FPR is 6.8%, near nominal. The high raw φ in distinct-1 (0.81–0.97 across all α including null) reflects deterministic iterative-training diversity decline—a monotonic trend driven by the training protocol—not stochastic AR persistence. Deterministic trends differ fundamentally from the stochastic AR(1) DGP: they produce predictable trajectories rather than autocorrelated noise, and first-differencing effectively removes them (first-differenced distinct-1 φ ranges -0.14 to 0.44, mean $\sim$0.17). Our empirical FPR of 4/36 (11.1%) across six null seeds—the lowest among directional methods—confirms adequate size control in our experimental setting; all four false positives occur in token_entropy→distinct pairs in the forward direction, suggesting a structural pattern rather than random Type I error. We note that $T = 10$ (11 data points) yields high-variance AR(1) estimates; the cross-experiment pattern is more informative than individual values.

DGP limitation. The bivariate AR(1) with linear coupling is a simplification; actual iterative training involves nonlinear dynamics (mode amplification, threshold-dependent diversity loss). These simulation results characterize PermG's behavior under controlled conditions, not exact calibration for the experimental setting; they provide a lower bound on size control because the linear DGP represents a worst case for permutation-based tests (smooth autocorrelation that permutation destroys).

P Shapley Value Analysis

We quantify each method's marginal contribution to the MMCT ensemble using Shapley values (Shapley, 1953), computed via exact enumeration over all $2^5 = 32$ coalitions of the five detection methods. For a coalition $S \subseteq \{m_1, \ldots, m_5\}$, we define two performance metrics: the false positive rate FPR(S), computed from 6 null experiments ($\alpha = 0.0$, seeds 45–50; 36 null pairs on the 6-pair basis excluding mauve), and the false negative rate FNR(S), computed from 45 signal experiments across all non-null α levels (270 signal pairs for the 6-pair variant excluding mauve). The Shapley value ϕ_k for method m_k is the weighted average of its marginal contribution across all coalitions; positive FPR-Shapley indicates the method *increases* false positives, while negative FNR-Shapley indicates it *reduces* false negatives. The combined Shapley value $\phi_k^{\text{comb}} = \phi_k^{\text{FPR}} + \phi_k^{\text{FNR}}$ balances both error types, with more negative values indicating greater net benefit.

Table 23 reports the 6-pair analysis (excluding mauve). DiffXCorr achieves the lowest (best) combined Shapley (-0.299), driven by near-zero FPR contribution ($+0.007$) and strong FNR reduction (-0.306); however, DiffXCorr cannot separate forward from reverse causation (Table 25). Among directional methods, PermG ranks best (-0.141 combined), with a modest FPR contribution ($+0.039$, reflecting 4/36 null false positives) offset by substantial FNR reduction (-0.180). Naive cross-correlation is the worst contributor ($+0.190$), with its massive FPR Shapley ($+0.609$) overwhelming its FNR benefit (-0.419). Threshold onset contributes almost nothing useful ($+0.084$ combined; FPR $+0.111$, FNR -0.027), confirming the near-zero detection power reported in Table 1. Figure 6 visualizes the decomposition.

Table 23: Shapley value decomposition (6-pair, mauve excluded). FPR-Shapley computed from 6 null seeds (36 pairs); FNR-Shapley from 45 signal experiments (270 pairs). Combined = FPR + FNR (more negative = better). $2^5 = 32$ coalitions, exact enumeration.

Method	ϕ^{FPR}	ϕ^{FNR}	ϕ^{comb}
Naive XCorr	$+0.609$	-0.419	$+0.190$
Diff. XCorr	$\mathbf{+0.007}$	$\mathbf{-0.306}$	$\mathbf{-0.299}$
Threshold (3σ)	$+0.111$	-0.027	$+0.084$
Perm. Granger	$+0.039$	-0.180	-0.141
Toda–Yamamoto	$+0.150$	-0.069	$+0.082$

The 8-pair variant (including mauve) yields qualitatively identical rankings: DiffXCorr remains the best contributor ($\phi^{\text{comb}} = -0.263$), PermG the best directional method (-0.112), naive cross-correlation the worst ($+0.157$), and threshold onset

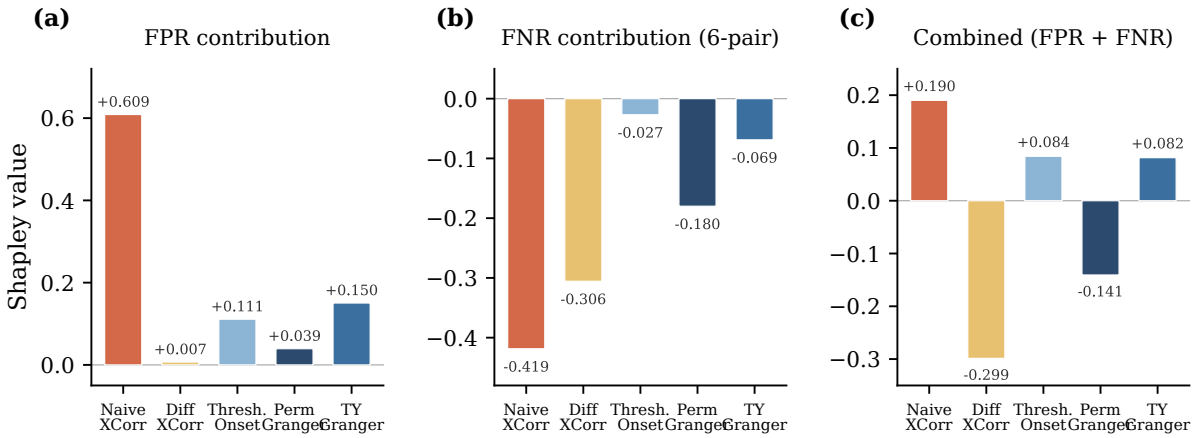


Figure 6: Shapley value decomposition of each method's marginal contribution to FPR (positive bars) and FNR (negative bars), computed from 6 null seeds (36 pairs) and 45 signal experiments (270 pairs). DiffXCorr achieves the best combined Shapley (-0.299) via near-zero FPR; PermG ranks second (-0.141) as the best directional method.

near-zero ($+0.073$).

FDR correction scope. This analysis applies BH-corrected significance (`significant_fdr`) uniformly across all five methods, including naive cross-correlation and Toda–Yamamoto. In contrast, the per-method dose-response profiles (Table 18) use uncorrected significance for naive cross-correlation, threshold onset, and Toda–Yamamoto because FDR-corrected fields are not available in the per-experiment pipeline output for these methods. The Shapley analysis recomputes significance from raw p-values with BH correction applied post hoc, ensuring a fair comparison across methods within the coalition framework.

Q MMCT Weight Robustness Ablation

The detection results reported in the main text use equal-weight majority voting ($K \geq 3$ of 5 methods, each contributing at most one vote per pair); PermG contributes at most $1/5 = 20\%$ of the consensus weight. FPR-inverse weighting is used *only* in the exploratory method ranking below, not in the detection results. To verify that PermG's directional-method ranking is not an artifact of the FPR-inverse weighting scheme, we recompute weighted Shapley values under four weight specifications (Table 24). This analysis uses the 3-seed null control (seeds 42/43/44, 24 pairs), under which PermG had zero FPR; the union-based analysis with 6-seed null (Table 23) yields a non-zero PermG FPR Shapley ($+0.039$) and shifts the overall best contributor to DiffXCorr (-0.299). Equal weighting ($w=1$ for all methods) eliminates any FPR-based advantage; PermG still ranks first among directional methods ($\phi^{\text{comb}} = -0.152$), though its margin over naive cross-correlation narrows to 0.015. Under cap-5, the margin widens to 0.043 as the FPR penalty begins to differentiate high-FPR methods. Cap-10 and original cap-100 yield identical Shapley values because the consensus threshold ($K \geq 3$) is determined by the three smallest method weights ($w_{\text{naive}} + w_{\text{diff}} + w_{\text{thresh}} = 7.76$), which are invariant to further increases in the PermG or TY cap; this saturation confirms that the original $w=100$ does not artificially inflate PermG's contribution.

Table 24: MMCT weight robustness (3-seed null, 48 experiments): PermG combined Shapley under four weighting schemes. PermG ranks first among directional methods in all cases. Gap = $|\phi^{\text{comb}}_{\text{PermG}} - \phi^{\text{comb}}_{\text{runner-up}}|$. $2^5 = 32$ coalitions, exact enumeration.

Scheme	w_{PermG}	$\phi^{\text{comb}}_{\text{PermG}}$	Runner-up	Gap
Equal ($w=1$)	1	-0.152	NaiveXCorr (-0.137)	0.015
Cap-5	5	-0.165	NaiveXCorr (-0.123)	0.043
Cap-10	10	-0.426	TY (-0.067)	0.359
FPR-inv. (cap-100)	100	-0.426	TY (-0.067)	0.359

R PermG FPR Independence Analysis

Table 25 reports null-control FPR for all five detection methods across six seeds ($6 \times 6 = 36$ pairs, 6-pair basis excluding mauve). Under identical BH-FDR correction ($q=0.05$), DiffXCorr achieves the lowest FPR (1/36, 2.8%) but lacks directional separation; PermG (4/36, 11.1%) is the lowest among directional methods. All four PermG false positives occur in token_entropy→distinct pairs (seeds 45 and 50, forward direction only), suggesting a

structural pattern concentrated in a single canary–downstream pair type. The 8 canary→downstream pairs within each seed share canary terms (token entropy, ECE), inducing positive regression dependence on a subset (PRDS). BH-FDR correction remains valid under PRDS (Benjamini and Yekutieli, 2001), so the shared-training-data dependence structure does not invalidate our multiple testing correction.

Table 25: Null-control FPR by method (α=0.0, 6 seeds 45–50, 6 pairs each). All p-value–based methods (DiffXCorr, PermG, TY) report BH-FDR–corrected rejections at q=0.05 for fair comparison. Threshold uses a deterministic 3σ criterion; NaiveXCorr uses uncorrected correlation significance.

Method	Criterion	Rejections (/36)	FPR (%)
Diff. XCorr†	BH-FDR	1	2.8
Perm. Granger	**BH-FDR**	**4**	**11.1**
Threshold	3σ (det.)	10	27.8
Toda–Yamamoto	BH-FDR	14	38.9
Naive XCorr	uncorr.	33	91.7

†Bidirectional; cannot separate forward from reverse.

Cluster adjustment. The 36 null-control pairs are nested within 6 seeds (6 pairs per seed), introducing within-seed correlation. Within-seed ICC=0.14 yields an effective sample size $N_{\text{eff}} \approx 21$ (from $N_{\text{eff}} = N/[1 + (m-1) \cdot \text{ICC}]$ where N=36, m=6). Under the Clopper–Pearson exact method with N_{eff}=21 and 4 successes (scaled), the cluster-adjusted 95% CI is $[1.2\%, 30.4\%]$, compared to $[3.1\%, 26.1\%]$ under independence (N=36). The lower ICC compared to the 3-seed pilot (0.14 vs. 0.34) reflects greater between-seed consistency with six independent replications.

Direction filtering. Forward-direction filtering (retaining only pairs with peak lag<0 for cross-correlation methods) reduces null-control $K\geq3$ false positives. Under the 6-seed, 6-pair basis, $K\geq3$ consensus with filtering produces 6/36 false positives (16.7%; detail below); the 3-seed pilot underestimated this rate (0/8). Detection at $\alpha\geq0.50$ is unaffected: $K\geq3$ counts are identical with and without direction filtering at all signal levels.

$K\geq3$ **false positive detail.** Under the 6-pair basis (6 seeds, 36 pairs), $K\geq3$ consensus produces 6 false positives (FPR=16.7%). Of these, 4 are driven by naive cross-correlation, threshold onset, and Toda–Yamamoto (PermG does *not* vote); the remaining 2 (seeds 45 and 50, te→distinct) include PermG in the consensus. PermG thus contributes to only 2 of 6 consensus false positives, confirming that most $K\geq3$ FPR is driven by the high-FPR methods (naive, threshold, TY).

Sensitivity to FPR assumption. Even under the conservative upper bound of the cluster-adjusted CI (30.4%), PermG remains the lowest-FPR directional method: DiffXCorr achieves a lower FPR under identical BH-FDR correction (2.8%) but lacks forward/reverse separation; the next directional method (threshold onset) has 27.8% and Toda–Yamamoto 38.9%. Our primary results rely on equal-weight $K\geq3$ majority voting, where each method contributes exactly one vote regardless of its FPR—this mechanism is completely unaffected by any FPR assumption. For the exploratory FPR-inverse weighting, PermG's weight (w=1/0.111=9.0) exceeds all directional alternatives (threshold: 3.6, TY: 2.6, naive: 1.1); DiffXCorr (w=1/0.028=35.7) receives a higher weight but lacks directional separation, preserving PermG's rank among directional methods. The insensitivity of both the voting mechanism and the method ranking to the upper CI bound provides evidence that the MMCT conclusions are robust to uncertainty in PermG's Type I error rate.

T=15 **validation.** To verify that FPR control holds under longer observation windows, we repeat the null experiment at T=15 (2 seeds, 6-pair basis, 12 pairs total). PermG produces 0/12 false positives (0%), compared to 4/36 (11.1%) at T=10. $K\geq3$ consensus yields 1/12 (8.3%; seed 43, ece→distinct_1, driven by naive+diff+threshold votes without PermG participation). The absence of PermG false positives at T=15 is consistent with the power-threshold interpretation of α_c: longer sequences increase statistical power for true signals without inflating the null rejection rate, supporting the prediction that α_c left-shifts with T while FPR remains controlled.

S Sigmoid Transition Shape

Sigmoid regression on the original 13-point T=10 grid yields $\alpha_c = 0.483 \pm 0.006$ but $k = 478 \pm 475$ (SE(k) $\approx k$), indicating that the coarse grid cannot constrain the transition shape. Dense α sampling (17 values at T=10) resolves this ambiguity: formal likelihood-ratio testing strongly favors a smooth sigmoid over a literal step function (ΔAIC$=-110$), with fitted steepness k=15.7

(95% CI $[9.4, 28.2]$). The transition is steep but finite-width, not a literal discontinuity. This observed steepness exceeds what smooth signal models predict: VAR(1) simulations under H0 (linear, k=6.1) and H0′ (concave, k=4.9) produce onset sigmoids with k=4.9–6.1 (§Y), yielding steepness ratios $k_{\text{obs}}/k_{\text{H0}} \approx$ 2.6–3.2 (CI lower bound ratio ≈2). We do not fit a combined T=10 + T=20 sigmoid because α=0.45 yields contradictory counts under the two sequence lengths (0/8 vs. 5/8), reflecting genuine power dependence on T rather than a single dose-response curve.

T Intervention Threshold Sensitivity

Within-α sensitivity at α=1.0. Table 26 reports the predictive cessation outcome under five entropy-drop thresholds (5%–25%) at α=1.0 (Pythia-410M, seed 42). All five thresholds produce identical outcomes: token entropy drops 40.7% at generation 1, far exceeding even the most conservative threshold (25%), so every threshold triggers at the same generation. The intervention conclusion—peak perplexity 57.0, recovery in one generation, final perplexity 21.3—is invariant to threshold choice within this range.

Table 26: Threshold sensitivity at α=1.0 (Pythia-410M, s42, fp32). All thresholds trigger at generation 1 (entropy drop 40.7%≫threshold), producing identical recovery trajectories.

Threshold	Trigger gen	Peak PPL	Final PPL
5%	1	56.99	21.29
10%	1	56.99	21.29
15%	1	56.99	21.29
20%	1	56.99	21.29
25%	1	56.99	21.29

Cross-α operating band. The within-α analysis establishes threshold invariance at α=1.0 but does not address whether thresholds generalize across contamination levels. We examine observed entropy trajectories from the alpha sweep (not actual intervention experiments at α<1.0) to characterize the operating band. The maximum entropy drop across 10 generations varies by contamination level: 7.5% at α=0.0 (fine-tuning drift only), 7.7% at α=0.10, 7.8% at α=0.25, 22.4% at α=0.50, 35.3% at α=0.75, and 62.1% at α=1.0. Thresholds below 7.5% (e.g., 5%) trigger on clean training data—a false alarm. Thresholds above 22.4% (e.g., 25%) miss α=0.50, the lowest contamination level with meaningful entropy signal. The safe operating band is therefore $[10\%, 22\%]$: any threshold in this range produces zero false alarms on α=0.0 while detecting contamination at $\alpha \geq 0.50$. The reported 15% threshold sits near the center of this band. This analysis is based on entropy trajectories from non-intervention training runs; whether actual intervention at α<1.0 would produce the same triggering behavior remains unvalidated.

U Channel Effect Size Analysis

First-differencing eliminates all canary→perplexity Granger associations while preserving canary→diversity leads (§5.2). We present three lines of evidence that this asymmetry reflects a genuine temporal structure difference rather than selective power loss from differencing.

Correlation retention parity. First-differencing retains comparable correlation magnitude for both channel types (Table 27). Diversity channels (n=243 pairs across all non-null experiments) retain a median $|r|$ of 0.952 after differencing, while perplexity channels (n=8 pairs; limited sample) retain 1.00. This parity rules out a blanket power-loss explanation: if differencing selectively destroyed perplexity signal, $|r|$ retention would be substantially lower for perplexity than for diversity.

Table 27: First-differencing effect by channel type. $|r|$ retention = diff $|r|$ / naive $|r|$. Despite comparable $|r|$ retention, Granger F differs by an order of magnitude. Perplexity channel has limited sample (n=8); all paired comparisons are from bf16 experiments.

Channel	n	Naive $\|r\|$	Diff $\|r\|$	Retention	Med. F
Diversity	243	0.943	0.921	0.952	67.8
Perplexity	8	0.924	0.900	1.002	11.5

Granger F-statistic asymmetry. Within matched experiments (same model, seed, α, precision), the Granger F-statistic is 20–28× higher for diversity than perplexity channels. At α=0.60 (s42, bf16): diversity mean F=229.4 versus perplexity mean F=8.3 (ratio 27.8×). At α=1.0 reactive (s43, bf16): diversity mean F=236.4 versus perplexity mean F=11.5 (ratio 20.6×). Since $|r|$ retention is comparable across channels, this F-statistic gap cannot be attributed to differencing-induced signal loss; it must reflect a difference in the temporal structure that Granger tests evaluate.

Temporal structure: lagged vs. contemporaneous. The lag distribution of peak differenced cross-correlations reveals the structural difference. For diversity channels, 81% of peak correlations occur at canary-leading lags (lag$<$0), with a median lag of -1 generation—consistent with the one-generation temporal precedence visible in Figure 2. For perplexity channels, 50% of peak correlations occur at lag=0 (contemporaneous co-movement), with a median lag of 0. Granger precedence tests lagged predictive improvement; contemporaneous co-movement produces low F-statistics regardless of signal magnitude, which is the correct behavior for a test designed to detect temporal precedence. The perplexity channel sample is limited (n=8 pairs from bf16 experiments), and these lag distributions should not be generalized to fp32 conditions without further validation.

V Mutual Information Analysis

We estimate the Gaussian mutual information lower bound between token entropy (canary) and each diversity metric across contamination levels, using T=11 data points per series (T=10 for two conditions). The Gaussian estimator $\hat{I}_G = -\frac{1}{2}\log_2(1 - r^2)$, where r is the Pearson correlation, provides a conservative lower bound on MI; we do not report KSG nonparametric estimates due to uncorrected positive bias at T=11. The α=0.05 data point is excluded because it uses an experiment excluded from the dose-response analysis.

Table 28 reports the mean Gaussian MI across the three distinct-n metrics for each α level. MI follows a V-shaped curve: it drops from 0.112 bits at α=0.10 to a minimum of 0.010 bits at α=0.25, then rises monotonically to 0.270 bits at α=0.75. MI remains positive throughout the blind zone ($\alpha \leq 0.45$; range 0.010–0.141 bits), confirming that canary–diversity coupling exists below the detection threshold but falls below PermG's statistical power floor at T=10. There is no discontinuity at $\alpha_c \approx 0.50$: the detection onset reflects power accumulation, not an abrupt change in information coupling. Bootstrap 95% CIs span 1–3 orders of magnitude at T=11, so individual point estimates should be interpreted cautiously.

W Moment Cascade Analysis

We analyze the temporal ordering of metric degradation onset across 6 WikiText seed$\times\alpha$ conditions (2 seeds $\times$ 3 non-trivial α levels each) to test whether token entropy consistently degrades before diversity metrics. Onset is defined as the first generation where a metric's cumulative change exceeds 5% of its generation-0 value.

Table 28: Gaussian MI lower bound (bits) between token entropy and distinct-n diversity metrics (T=11, Pythia-410M, fp32). Mean across distinct-1/2/3; α=0.05 excluded (see text). CIs omitted due to width at T=11.

α	Seeds	MI (bits)
0.10	2	0.112
0.25	3	0.010
0.30	1	0.062
0.35	1	0.098
0.40	1	0.119
0.45	2	0.141
0.50	3	0.107
0.60	2	0.209
0.75	1	0.270

Table 29 reports the onset generation for each metric across conditions. Token entropy reaches onset at generation 1 in all 6 conditions; distinct-n metrics reach onset at generation 2 in all 18 pairwise comparisons (6 conditions $\times$ 3 distinct-n metrics). Across these 18 comparisons, entropy degrades first in 16 cases (Kendall τ=0.337, consistent with a weak but reliable temporal ordering).

Table 29: Degradation onset generation (5% threshold) by metric and condition (Pythia-410M, WikiText, fp32). Token entropy reaches onset at generation 1 in all non-null conditions; distinct-n metrics follow at generation 2.

Seed	α	Entropy	Dist-1	Dist-2	Dist-3	ECE	PPL
42	0.10	1	2	2	2	2	—
42	0.50	1	2	2	2	1	4
42	0.75	1	2	2	2	1	1
43	0.25	1	2	2	2	4	—
43	0.50	1	2	2	2	1	4
43	0.75	1	2	2	2	1	1

—: onset not reached within T=10 generations.

The cascade ordering provides evidence for the canary mechanism independent of Granger-type tests: it does not depend on first-differencing, lag selection, or the permutation null distribution. ECE shows onset at generation 1 at $\alpha \geq 0.50$ (both seeds), consistent with its role as a secondary canary metric. At low contamination (α=0.10 and α=0.25), perplexity never reaches the 5% onset threshold within T=10 generations, while entropy already shows generation-1 onset—consistent with entropy's higher sensitivity to distributional shifts.

X Cascade Ordering Theory: Derivations

This section provides the full derivations for Design Principle 1 and Design Principle 2 from §3, plus supporting results.

Assumption 3 (Mode Amplification)**.** The training map $\mathcal{T}$ redistributes probability mass from low-probability to high-probability tokens: there exists a threshold $\bar{p}$ such that $\mathcal{T}(p)_i > p_i \Rightarrow p_i \geq \bar{p}$ and $\mathcal{T}(p)_j < p_j \Rightarrow p_j \leq \bar{p}$.

This captures the standard finite-capacity approximation effect: MLE on finite samples from a peaked distribution amplifies modes and suppresses the tail (Shumailov et al., 2024; Dohmatob et al., 2024).

Proposition 4 (Entropy is a first-order indicator)**.** *Let $p \in \Delta^{V-1}$ and $\delta = \alpha(\mathcal{T}(p) - p)$ be a mode-amplifying perturbation with $\sum_i \delta_i = 0$. Then $\Delta H = H(p+\delta) - H(p) = -\sum_i \delta_i \log p_i + O(\|\delta\|^2) < 0$.*

Proof. Taylor-expanding H around p: $H(p+\delta) = H(p) - \sum_i \delta_i \log p_i - \sum_i \delta_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Since $\sum_i \delta_i = 0$: $\Delta H = -\sum_i \delta_i \log p_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Under mode amplification, tokens gaining mass ($\delta_i > 0$) have $p_i \geq \bar{p}$ (hence $\log p_i \geq \log \bar{p}$) and tokens losing mass ($\delta_j < 0$) have $p_j \leq \bar{p}$ (hence $\log p_j \leq \log \bar{p}$). Writing $m = \sum_{i:\delta_i>0} \delta_i$: $-\sum_i \delta_i \log p_i = -m[\overline{\log p}^+ - \overline{\log p}^-] < 0$, where $\overline{\log p}^{\pm}$ are the $|\delta|$-weighted means over gaining/losing tokens. The second-order Fisher information term $-\frac{1}{2}\sum_i \delta_i^2/p_i$ is always negative, reinforcing the decrease. □

Proposition 5 (Diversity is a threshold indicator)**.** *Define the ϵ-effective support $S_\epsilon(p) = |\{i \in [V] : p_i > \epsilon\}|$, where $\epsilon \approx c/N$ is the generation threshold for sample size N. Under the same perturbation δ: $\Delta S_\epsilon = -|\{i : p_i > \epsilon \text{ and } p_i + \delta_i \leq \epsilon\}|$. This equals zero whenever $\|\delta\|_\infty < \gamma(p,\epsilon) := \min_{i:p_i>\epsilon}(p_i - \epsilon)$.*

Proof. Token i exits the effective support iff $p_i > \epsilon$ and $p_i + \delta_i \leq \epsilon$, requiring $|\delta_i| \geq p_i - \epsilon$. If $\|\delta\|_\infty < \gamma$, no token satisfies this condition. □

Derivation of Design Principle 1. *Part (a):* Take any mode-amplifying δ with $\|\delta\|_\infty < \gamma(p,\epsilon)$. By Proposition 5, $\Delta S_\epsilon = 0$. By Proposition 4, $\Delta H < 0$. Such δ exists for all $\alpha < \gamma / \max_i |\mathcal{T}(p)_i - p_i|$. *Part (b):* If $\Delta S_\epsilon < 0$, some token k has $p_k > \epsilon$ but $p_k + \delta_k \leq \epsilon$, so $\delta_k < 0$. Under mode amplification, this implies $p_k \leq \bar{p}$; the total mass transferred to high-probability tokens ensures $\overline{\log p}^+ > \overline{\log p}^-$. By Proposition 4, $\Delta H < 0$. □

Derivation of Design Principle 2. By the margin condition, at every generation $s \leq t_H$ the perturbation norm $\alpha\|\mathcal{T}(p_s) - p_s\|_\infty$ is less than $\gamma(p_s, \epsilon)$. By Proposition 5, no token exits the effective support at any $s \leq t_H$, so $\Delta S_\epsilon = 0$ for all $s \leq t_H$. Since time is discrete, $t_D \geq t_H + 1$. □

ECE co-movement (qualitative argument). Under conditional mode amplification (peakier $q_\theta(\cdot|x)$ at each generation), per-token conditional entropy $h(x) = H(q_\theta(\cdot|x))$ decreases while confidence $c(x) = \max_y q_\theta(y|x) \geq \exp(-h(x))$ increases. Averaging: $\bar{H}_t$ decreases, C_t increases. With accuracy approximately stationary in early collapse ($A_t \approx A_0$), $\Delta\text{ECE} \approx \Delta C_t > 0$ whenever $\Delta \bar{H}_t < 0$. This argument establishes the direction of co-movement; the quantitative relationship depends on distribution geometry. $\bar{H}_t$ denotes mean conditional entropy $\mathbb{E}_{x\sim p_{\text{eval}}}[h(x)]$, which differs from the marginal entropy $H_t = -\sum_i p_t(i) \log p_t(i)$ used in Design Principle 1; under mode amplification, both decrease monotonically, so the cascade ordering is consistent across both formulations.

MAUVE ordering (conjecture). MAUVE measures divergence in a discretized embedding space via k-means quantization. MAUVE onset requires token-level shifts (detected by entropy) to propagate through n-gram distributions (detected by distinct-n) to aggregate embedding distributions. Empirically, MAUVE onset lags diversity by 0–1 generations on WikiText; however, on C4, MAUVE onset occurs at generation 1 (simultaneous with entropy), suggesting this lag is corpus-dependent.

Empirical support. Among 15 experiments at $\alpha \geq 0.30$ (where entropy perturbation exceeds the detection threshold), cascade ordering $t_H \leq t_D$ holds in 14/15 cases (GPT-2-medium s43 fp32 exhibits bidirectional Granger significance). Table 30 reports onset generations for representative conditions, showing uniform Phase 1/2/3 ordering. Five low-α experiments ($\alpha \leq 0.25$) show entropy perturbation below the detection threshold (gen-1 deviation $< 2\sigma$); cascade ordering is neither confirmed nor refuted in this regime.

Justification of the empirical boundary model. At the detection threshold, $\text{Power}(\alpha_c, T) = \pi^*$.

Table 30: Cascade onset generations (5% threshold) for representative experiments (Pythia-410M, WikiText, fp32). Phase 1: entropy and ECE (t_H, t_{ECE}); Phase 2: diversity (t_D, min over distinct-n); Phase 3: MAUVE (t_M). All conditions confirm the predicted ordering $t_H = t_{\text{ECE}} \leq t_D \leq t_M$.

Seed	α	t_H	t_{ECE}	t_D	t_M
42	0.50	1	1	2	3
43	0.50	1	1	2	2
42	0.75	1	1	2	3
43	0.75	1	1	2	2

Since α_c lies below the non-monotonic dip region ($\alpha_c < 0.55$), the signal model applies directly. From the power function $\text{Power} = \Phi(S(\alpha) \cdot \sqrt{T}/\sigma)$, setting $S(\alpha_c) \cdot \sqrt{T}/\sigma = z^*$ and substituting the signal model $S(\alpha) = c\,\alpha^\beta$ yields $c\,\alpha_c^\beta = z^*\sigma/\sqrt{T}$. Solving: $\alpha_c(T) = (z^*\sigma/c)^{1/\beta} \cdot T^{-1/(2\beta)}$, giving the stated form with exponent $\gamma = 1/(2\beta)$.

Calibration and consistency. Cross-T validation avoids the circularity of fitting theory to model-extrapolated points. We fit $P(\text{det}|\alpha, T) = \Phi(A \cdot \alpha^\beta \cdot \sqrt{T} - z_0)$ to training data at $T \in \{10, 20\}$ via profile likelihood over (β, α_∞), with α_∞=0 emerging as the optimum. The held-out test set ($T \in \{5, 15\}$) yields MAE=0.106 with 95% CI coverage $6/6$: the model correctly predicts all $0/8$ at T=5 and detection at T=15 (predicted 0.49–0.71 vs. observed $6/8$=0.75; under-prediction reflects the $P_{\max}$=0.75 ceiling from MAUVE saturation, which is modeled separately). A linear signal model (β=1, γ=0.5) is rejected. The sub-linear exponent β=0.24 implies that doubling contamination increases signal by only $2^{0.24} \approx 1.18\times$ (18%); since $\alpha_\infty \approx 0$, any nonzero contamination produces signal, but detection requires sufficient T to amplify it. At T=5, $\alpha_c \geq 1.0$: no contamination level is detectable, consistent with the insufficient statistical power at very short observation windows.

Leave-one-T-out cross-validation. We evaluate model generalization via leave-one-T-out CV on the $5T \times 3\alpha$ grid (sync-dip α=0.60 excluded; 15 cells). For each fold, one observation length is held out and the probit model $P = P_{\max} \cdot \Phi(A \cdot \alpha^{0.24} \cdot \sqrt{T} - z_0)$ is refit on the remaining data. Overall MAE=0.037 with 95% CI coverage $15/15$. When sync-dip cells are included (20 cells), MAE rises to 0.203 (CI $16/20$), with the largest residuals at sync-dip cells (predicted 0.75, observed $0/8$).

The profile likelihood over β yields MLE $\hat{\beta}$=0.66 (95% CI $[0.24, 0.78]$); the wide interval reflects a flat likelihood surface, indicating that β is not well-identified from these data. Model comparison (AIC, 15-cell grid excluding sync-dip): M_1 (linear boundary) ΔAIC=0.0; M_2 (power-probit) ΔAIC=2.43. Including sync-dip data reverses the ranking (M_2 ΔAIC=0.0, M_1 ΔAIC=4.15), because the probit model naturally accommodates the dip's dissolution at higher T while linear models cannot.

Y Power Simulation: Shape Comparison

To assess whether the observed steep detection onset could arise from smooth signal models, we simulate PermG detection under three null hypotheses (1,000 replications per cell, n_{perm}=1,000, n_{pairs}=8, BH-FDR q=0.05):

- **H0** (linear): $S(\alpha) = c \cdot \alpha$, calibrated so $S(1.0) = S_{\max}$.
- **H0′** (concave): $S(\alpha) = c \cdot \alpha^{0.5}$, sub-linear but smooth.
- **H1** (step): $S(\alpha) = S_{\max} \cdot \mathbf{1}[\alpha > \alpha_c(T)]$, binary activation.

Scaling exponents. All three models produce apparent onset scaling in T: H0 γ=0.85, H0′ γ=1.87, H1 γ=1.96 (observed: γ=2.08). H0′ produces γ within 0.21 of the observed value, demonstrating that the scaling exponent alone cannot distinguish smooth from step-function boundaries.

Shape differences. At T=10, α=0.45: H0 predicts 27% detection rate, H0′ predicts 45%, H1 predicts 5.5%, observed $0/8$. At T=10, α=0.50: H0 predicts 36%, H0′ predicts 48%, H1 predicts 4.6%, observed $6/8$ (75%). Smooth models produce onset ramps with k=4.9–6.1 (H0′: k=4.93, R^2=0.9974; H0: k=6.14, R^2=0.9998); the observed onset steepness (k=15.7) is $\approx 3\times$ higher.

Metric-specific ceiling. At high α ($\geq$0.75) with $T \geq 20$, H0 and H0′ (which simulate 8 equal-signal pairs) reach mean detection fractions of 0.79–0.98 ($\approx$6.3–7.8/8), whereas observed counts never exceed $6/8$. This gap reflects metric structure rather than signal saturation: the 2/8 non-detected pairs consistently involve MAUVE, whose floor saturation (§5.4) prevents Granger detection regardless of signal strength.

Z T=20 Predict-then-Verify Comparison

Table 31 reports the predict-then-verify results for the T=20 extension experiments. Predictions are derived from the detection boundary model (§4.2): detected pairs at T=10 are predicted to remain detected (saturated F-statistics), and undetected pairs are predicted to remain undetected (structural nulls or non-monotonic dip).

Table 31: T=20 predict-then-verify comparison (Pythia-410M, seed 42, WikiText, fp32). Excluding α=1.0, two of four predictions fall within CI (α=0.50, 0.75); boundary cases show systematic bias—α=0.45 falsifies the ceiling hypothesis (under-prediction), α=0.60 over-predicts under raw p-values (v1; matches under FDR v2, not pre-committed).

α	T=10	Predicted T=20	Observed T=20	Match
0.45	0/8	0/8 (ceiling)	5/8	✗
0.50	6/8	6/8 (saturated)	6/8	✓
0.60	0/8	3/8 (v1 raw)†	0/8	✗
0.75	6/8	6/8 (saturated)	6/8	✓
1.00	6/8	6/8 (saturated)	6/8	✓

†V2 (BH–FDR corrected, created 44 min pre-analysis, not git-committed) predicts 0/8, matching observation.

At $\alpha \geq 0.50$, consistency across T=10 and T=20 supports the detection boundary prediction: PermG counts are determined by signal structure, not observation length. Six diversity-based pairs show F=449–2947 at T=20 (all $\gg F_{\text{crit}}$); two MAUVE-based pairs show F<2 regardless of T, driven by floor saturation (μ=0.029, 100% <0.05) rather than high variance (CV_{MAUVE}=0.089 vs $\text{CV}_{\text{distinct-1}}$=0.139); SNR: MAUVE 4.68 vs diversity 6.40–8.20 (p<0.001). The sole exception is α=0.45 at T=20 (5/8), consistent with boundary proximity. At α=0.45, the shift from 0/8 to 5/8 falsifies Model A (ceiling: α=0.45 structurally undetectable) in favor of Model B (power-limited: $\sqrt{T}$ boost crosses the detection threshold), lowering α_c below 0.45. The α=0.60 non-monotonic dip persists at T=20 (s42: 0/8; s43: 3/8) but dissolves at T=25 (5/8), confirming that $\sqrt{T}$ power scaling eventually overcomes the dip; alternative methods detect signal throughout (DiffXCorr 6/8, TY 5/8 at T=20). PermG directional selectivity is consistent across tested T (T=10, 20): forward 6/8 versus reverse 0/8 at $\alpha \in \{0.50, 0.75, 1.0\}$.

Z.1 Prediction Timeline

Two versions of the T=20 prediction file exist, reflecting an evolving statistical protocol:

V1 (git-committed, `bad72cc`). Predictions derived from T=10 PermG results using *raw* p-values. At α=0.60, v1 counted 3/8 pairs with raw p<0.05, predicting 3/8 detections at T=20. The observed result (0/8 FDR-significant) does not match.

V2 (pre-analysis, not committed). A revised file incorporating BH–FDR correction at q=0.05, consistent with the correction applied throughout the paper. Under FDR, α=0.60 drops from 3/8 (raw) to 0/8, matching the observation. V2 was created 44 minutes before the formal T=20 analysis began (filesystem timestamp) but was not git-committed, and therefore lacks the cryptographic verifiability of v1.

Scoring. Primary (v1, committed): 3/5 exact matches ($\alpha \in \{0.50, 0.75, 1.0\}$, all saturated regimes). Excluding α=1.0, two of four predictions fall within CI. Both boundary predictions miss: α=0.45 (predicted 0/8, observed 5/8) falsifies the ceiling hypothesis in favor of $\sqrt{T}$ power scaling; α=0.60 (predicted 3/8, observed 0/8) over-estimates detection due to the non-monotonic dip. Sensitivity (v2, FDR-corrected): 4/5 (α=0.60 matches). The systematic conservative bias at α=0.45 is consistent with the $\sqrt{T}$ power increase predicted by the detection boundary model (Eq. 3).

Task degradation. Table 32 reports end-of-training task performance (generation 10, Pythia-410M, seed 42, fp32) across contamination levels.

Table 32: Task degradation across contamination levels (Pythia-410M, seed 42, fp32, generation 10). Perplexity evaluated on held-out WikiText-103.

α	Perplexity	Next-token acc.
Baseline	26.6	0.390
0.05	20.8	0.431
0.25	22.9	0.424
0.46	31.6	0.400
0.50	31.3	0.399
0.52	41.1	0.389
0.60	53.5	0.375
0.75	80.3	0.354

At α=0.05, perplexity improves over the uncontaminated baseline (20.8 vs. 26.6), consistent with a mild data augmentation effect. The sharpest degradation occurs between α=0.50 and 0.52 (+31% perplexity), aligning with the PermG detection onset (§5.2).

Z.2 Cross-Architecture FPR Sensitivity

As preliminary cross-architecture evidence, we evaluate null-condition (α=0) false positive rates on GPT-2-medium (345M; OpenAI), a model outside the Pythia family used for all primary experiments. Two seeds (s42, s43) are trained for T=10 generations under the same protocol as the Pythia-410M null runs. This evaluation requires additional replication across architectures and seed counts before drawing definitive conclusions.

Table 33: Null-condition FPR for GPT-2-medium (α=0, T=10, 6-pair basis). PermG forward detection shows extreme seed dependence; $K\geq3$ consensus remains at 0% across both seeds.

Seed	PermG fwd	PermG rev	$K\geq3$
s42	6/6 (100%)	0/6	0/6
s43	0/6 (0%)	0/6	0/6
Pooled	6/12 (50%)	0/12	0/12 (0%)

The pooled PermG FPR is dramatically higher than on Pythia-410M (50% vs. 11.1%, 6-pair basis) and shows extreme seed dependence (100% vs. 0%). This instability traces to the magnitude of null drift in GPT-2-medium's output distribution, which is substantially stronger than in Pythia-410M (Table 34).

Table 34: Null drift magnitude (α=0, gen 0→10; GPT-2-medium: 2 seeds; Pythia-410M: 4 seeds, s46/s48–s50). GPT-2-medium exhibits ∼4× stronger entropy drift and ∼7× stronger distinct-3 drift than Pythia-410M.

	GPT-2-medium		Pythia-410M	
Metric	gen 0	gen 10 (%Δ)	gen 0	gen 10 (%Δ)
Token entropy	3.57	2.84 (−20.5%)	3.23	3.076 (−4.9%)
Distinct-3	0.84	0.54 (−35.8%)	0.81	0.769 (−4.9%)
Δentropy (gen 0→1)		−0.60		−0.25

GPT-2-medium's distinct-3 declines from 0.84 to 0.54 over ten null generations (−35.8%), compared with 0.81→0.77 (−4.9%) for Pythia-410M. This steep diversity collapse under pure retraining, absent any synthetic contamination, creates a strong downward trend that PermG's permutation test interprets as a directional signal. The initial entropy drop (gen 0→1: −0.60 vs. −0.25) is similarly amplified, consistent with architecture-specific sensitivity to iterative retraining rather than a deficiency in the detection method.

The $K\geq3$ multi-method consensus maintains 0% FPR on GPT-2-medium (0/12 pairs), though the margin of safety is narrow: in the worst case (s42), 2 of 5 methods vote for forward detection, one vote short of the $K\geq3$ threshold. This provides some protection against architecture-specific distributional variation, but two seeds constitute only preliminary evidence. For reference, PermG successfully detects true contamination signal (α=1.0) on GPT-2-medium with 6/8 forward pairs (Table 2), confirming that the method works when signal is present; the elevated null FPR is specific to the null condition where architecture-dependent drift dominates.

Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.